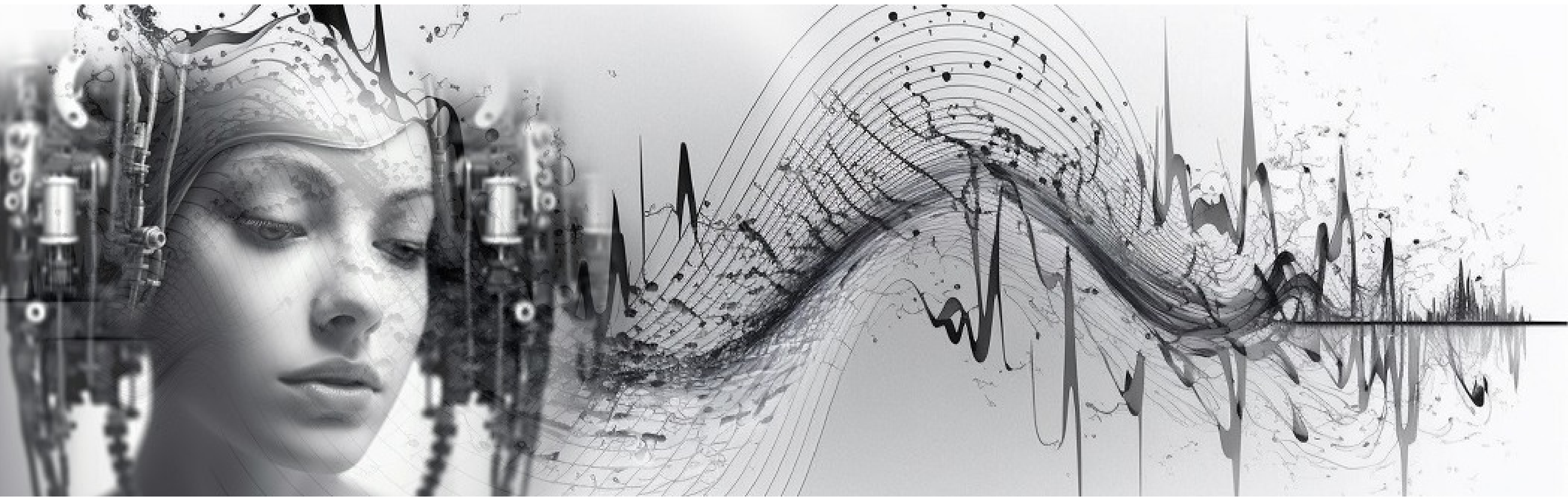

ElectroMUSE

Proceedings of the Australasian Computer Music Conference 2023

Hosted by the University of New England — Sydney campus, Parramatta, NSW

9–11 October 2023

Australasian Computer Music Association

Proceedings of the Australasian Computer Music Conference 2023

Australasian Computer Music Conference 2023 — *ElectroMUSE*

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ACMA acknowledges the Traditional Custodians of the lands on which we gather to share, play and learn, and pays respect to Elders past and present, and to all Aboriginal and Torres Strait Islander peoples.

http://computermusic.org.au

Contents

About ACMC 2023

The University of New England hosted the 2023 Australasian Computer Music Association Conference and Concert Series. ACMC is the annual gathering of the Australasian Computer Music Association where researchers and artists share and discuss emerging research and innovative practice in computer/electronic/digital music. ACMC is a festival of performances, installations, workshops, and tutorials that inspire, challenge, and showcase the emerging work of leading researchers and artists in this rapidly growing community of practice.

- **Dates:** 9–11 October 2023
- **Location:** University of New England, Sydney campus, Parramatta, NSW
- **Theme:** ElectroMUSE
- **Venues:** 100 George St (papers, Level 4), 28 George St (UNE installations & concerts), PHIVE / 5 Parramatta Square, and Riverside Theatres — presented in partnership with Parramatta City Council and the Parramatta Lanes Festival.
- **Full papers (Chroma):** https://journal.computermusic.org.au/chroma/issue/view/3

2023 Conference Theme: ElectroMUSE

This year's ACMC theme was ElectroMUSE. The Muses of Ancient Greek mythology are nine goddesses who preside over the arts and science and were the source of inspirational fire for the artist. To muse on something is to be absorbed in thought and to deeply reflect. ACMC2023 absorbed participants in thinking deeply about music and its relationship to all things technological, digital, and electronic. We took inspiration from the technological muse of our age and engaged with an array of creative processes and musical outputs throughout the event.

Technology has long been an integral part of music making practice and now more than ever, is a source of inspiration and collaboration among musicians and artists working with sound.

Humans and machines continue to evolve, becoming ever more deeply entwined, with greater power than ever before, to impact our social and cultural fabric and alter the world. Technology offers great opportunity and promise but also brings challenges and many unknowns which can generate fear and distrust. As we navigate our rapidly changing technological world, we muse upon, explore, imagine, investigate, reflect, make, collaborate, and draw inspiration from the emerging technologies that are fast becoming the Muse of the modern age, inspiring our music and music making practices.

Each concert and gallery in the program was named for a "muse" of the electronic and technological age — Choreobot, Cybella, KleioData, Calligrapix, Lyrictron, EuterMechania, Astraline, CybErato, and Techalia.

Keynotes

Keynote 1 — Asphyxia: AMPLIO, A Project Revolutionising Music Accessibility for Deaf People

- Monday 9 October, 10.30–11.30am — 100 George St, Level 4
- Chair: Bridget Johnson. Auslan Translator: Bel Diggins

Abstract:

Asphyxia presents a talk on her ground-breaking work in developing access to music for Deaf people. Asphyxia explores how deaf and hearing brains are wired differently when it comes to music, how hearing aids and cochlear implants affect our enjoyment of music, and how a lack of

music experience can alter our preferences. Asphyxia's passion and commitment have led her to investigate the best ways for deaf people to enjoy music through the creation of Amplio, which aims to solve barriers to music access. Learn what it takes to make music more accessible for deaf people and how new technology can help to make this possible. During this presentation Asphyxia unravels the features, capabilities, and ambitions of Amplio, drawing comparison to the transformative impacts of platforms like Spotify. Amplio enables people across the hearing spectrum to feel, see, hear and customise music to their needs.

Bio:

Asphyxia is a Deaf artist, writer, musician and public speaker. Asphyxia is creating Amplio, which will be like Spotify for Deaf people. It is assumed that deaf people will listen to music made for hearing people, but Asphyxia is rebuilding it from the ground up, so that it can be seen, heard, felt and customised to an individual's range. Asphyxia is also the author of *Future Girl*, winner of both the Schneider Family and Readings YA Book Awards for best book for teens, and soon to be adapted for the screen. *Future Girl* has been translated into Korean and Italian. Her books, including the much loved junior fiction series, *The Grimstones*, have also been shortlisted for numerous other awards. Asphyxia has also been a circus performer and puppeteer, and recently performed songs she composed for *Stranger Than Usual* with a live band in Melbourne. Deaf since the age of three, Asphyxia learnt to sign when she was eighteen, which changed her life. She is now a Deaf activist, sharing details of Deaf experience and raising awareness of oppression of Deaf people and what we can do to change this. Her free online Auslan course at asphyxia.com.au has over 15,000 students.

Keynote 2 — Dr Richard Savery: Robotic Musicianship

- Wednesday 11 October, 3–4pm — 100 George St, Level 4
- Chair: Dr Roger Alsop
- *Robotic Musicianship: Music and Creativity for Improved Artificial Intelligence and Robotics*

Abstract:

Robotic musicianship is an emerging interdisciplinary field that brings together music, computer science, performance studies, artificial intelligence (AI), and mechatronics. This presentation offers insights into three cutting-edge projects focused on the ways in which studies of robotic musicianship can extend into general AI and the robotics industry. Project 1: Emotional Musical Prosody, explores the fusion of musicianship with robotics, exploring how embedding musical phrases into industrial robotics can enhance trust and collaboration between musicians and machines. Project 2: Interactive Rapping Robot, describes a system for an interactive rapping robot and the implications for expert human–robot interaction. Project 3: Robot Musical Practice, describes a new system for robotic musical practice and the implications for robot development.

Bio:

Dr Savery is a developer of artificial intelligence and robotics, using music and creativity as a medium to program better interactions, understandings and models. He is a Research Fellow at Macquarie University, developing new robotic musicians. Dr Savery completed a PhD in Music Technology (minor in Human Computer Interaction) at the Georgia Institute of Technology, graduating in 2021. His research was conducted in the Robotic Musicianship Lab with Gil Weinberg. Much of his PhD work was based on an $800,000 NSF-National Robotics Initiative grant that he received to develop forms of emotional musical prosody to increase trust in group human–robot collaboration. Prior to Georgia Tech, Richard worked as a freelance music technologist, composer and performer. He completed his Masters at the University of California, Irvine, in Integrated Composition, Improvisation and Technology. He has composed and orchestrated many video games, films and ads including *Fast Four* featuring Roger Federer, and worked for 18 years as a professional saxophonist, clarinetist and flutist. https://richardsavery.com/

Paper Sessions

Paper Session 1A — (Monday 9 October)

- 100 George St, Level 4. Chair: Mark Oliveiro

From disco to digital and beyond: artist–audience dynamics and evolving topographies of the dance floor (Michael Callander & David Haberfeld)

Abstract:

The spatial and cultural dynamic between electronic dance music (EDM) artists and audiences has evolved significantly since the Disco dance floors of the 1970s. Originally working as near-invisible brokers of recorded music for a dancing audience, post-Disco DJs became more prominent in the 1980s and 1990s, eventually commanding large-scale festival stages in the 21st century. Today, many DJs and live electronic acts pursue a performer–audience orientation that is more varied and more dynamic, shifting away from stage-based, dance-floor-facing spaces to occupy novel environments and pursue new approaches to staging and optics. Concurrently, modern technologies inform the realisation of new performance practices, while digital distribution channels facilitate audience connections far beyond traditional dance floors. After articulating a series of developments before and after the turn of this century, this paper discusses how the future relationship between EDM makers and consumers can be further transformed: creativity, technology and online communities will continue to overlap and intersect, facilitating new approaches to performance, staging, and audience orientation.

Bio:

Dr Mike Callander is a Lecturer in Music Industry at RMIT University. His research examines the intersections between music technology and DJ culture. When he's not teaching or researching, you'll find Mike on tour or in the studio with bands like The Avalanches and The Presets, or at Melbourne's infamous Revolver nightclub, where he has been a weekly resident since 2010. Mike is also an Ableton Certified Trainer, one of around 300 people worldwide endorsed to teach Live and Push, and one of only two in Australia to have both this accreditation and a PhD.

Dr David Haberfeld is an accomplished and awarded electronic dance music artist, academic, researcher and music technologist with diverse experiences as producer, composer, performer and DJ since the early 1990s. David has successfully led and responded to the rapid changes in the way we produce, perform, market, manage, and teach music for almost three decades, including how digitalisation has impacted audience engagement. His experiences are unique, combining academic, industrial, and creative practices with over 200 published music works. Under the artist moniker Honeysmack, he is recognised as someone who has significantly contributed to Australia's electronic dance music community for many years.

Projecting into the Future: Mapping out a Starting Place (Lisa Griffiths)

Abstract:

This talk provides an overview of the author's 'Projecting into the Future' honours project, comprising a 3-D mapping artwork, *Immersion*, accompanied by supporting research that explores the career pathways of emerging 3-D mapping artists. The research sheds light on how these artists can develop their practice, enhance their skills, and establish their unique artistic voice

and profile within this evolving field. To achieve these objectives, the author conducted a series of ethics-approved interviews with established artists in the 3-D mapping field. Through these interviews, the author explores the artists' career trajectories, examining the challenges they encountered and the strategies they employed to advance their practice. By identifying commonalities and unique aspects in these artists' experiences, the author collected data that was then coded and analysed as part of the research process. The presentation also discusses the author's own creative work, outlining the development process and the approaches undertaken to refine their skills, and the approaches to creating portable and achievable artwork with minimal budgetary support.

Bio:

Lisa Griffiths' interest in mixed-media projects and 3-D technology has led her to research the field of interactive projection mapping. She is currently completing an Honours degree at UNE under the supervision of Associate Professor Donna Hewitt. Classically trained in Melbourne, Lisa was a student at MLC in Victoria and was awarded a private music scholarship at age 12. As a performing artist and composer, Lisa has a 20-year career history as a full-time professional musician that encompasses performing at numerous national and international multicultural events. Lisa was signed by the former Australian Promotions manager of Polygram (Universal Records) in 2003. Highlights of her career include performances at The Hammerstein Ballroom at the Manhattan Centre in New York and The Sydney State Theatre.

Whisper, a web app for audience participation and multichannel diffusion (Treya Nash)

Abstract:

This paper presents a whispering-gallery-inspired web app called *Whisper*, designed for collaborative performance between audience cell phones and a performer. The audience record themselves whispering into their cellphones, then upload their audio samples to a server. A performer controls a separate webpage through which they diffuse the samples into multichannel space, using the JSAmbisonics library. The piece is intended to engage the audience and give them agency through the use of their voice and the choice of whether to upload or delete their recording. The soundworld of the piece is designed to create a highly intimate and immersive experience which plays with the intelligibility of speech. In the first section, audience members record themselves whispering lines of a poem. In the second section, audience members choose what they whisper, which is then obscured through granular processing using tone.js. The Whisper framework is designed to be adaptable for use in future performance pieces.

Bio:

Treya Nash is an English composer and creative coder based in Baton Rouge, Louisiana. Her areas of focus include chamber music, distributed performance, and electronic opera. Her work has been performed internationally by many renowned ensembles. She is currently pursuing her PhD at Louisiana State University, with Mara Gibson, Jesse Allison, and Steven David Beck. She has previously studied with Paul Koonce, Mark Engebretson, and Alejandro Rutty, and worked as Program Director for Charlotte New Music.

Paper Session 1B — (Monday 9 October)

- 100 George St, Level 4. Chair: Roger Alsop. Theme: Production

Exploring the Advancements in Digital Audio Effects and the Role of Artificial Intelligence in Audio Production (Wenzheng Cui)

Abstract:

This presentation discusses two key developments in the field of digital audio production: Automatic Audio Production Methods (AAPMs) and Intelligent Music Production (IMP). Automatic Audio Production Methods utilise artificial intelligence and machine-learning techniques to enable a more efficient and effective approach to audio production, while IMP involves the development of intelligent systems that can analyse input audio and make decisions based on machine-learning skills developed from human interaction. Despite the challenges that remain, such as interpretability and computational complexity, the recent development of digital audio production has provided more possibilities.

Bio:

Wenzheng Cui is a graduate from the Australian Institute of Music majoring in Audio, currently pursuing his Master's degree in Music Technology at New York University. As a student at the Australian Institute of Music, he achieved over 1200 recording hours as an approved audio student. He has designed detailed recording plans including signal flow, acoustic analysis, miking techniques, and floor plans, directed projects, collaborated with 500+ performers, songwriters, producers, and engineers from over 8 countries, and delivered over 300 mixes in various genres and audio formats. He was selected as the recording & mixing tutor at AES for the AES DE+I Mentorship Program 2023 Term 1. Wenzheng's research topics cover AI Music, Digital Signal Processing, Audio Engineering, and Algorithmic Music Production.

Implications of Modular Music Composition for Digital Environment as a Means of Creating Dynamic Musical Works (Samuel Lynch)

Abstract:

The potentials and affordances of dynamic music-making have imposed a new level of flexibility in the music that we create. Music can now be created to dynamically change and adapt in real-time as is commonly seen in videogames, and this extra level of flexibility and non-linearity in what was typically a linear medium inspires a new understanding of how music can exist and how it can be made. This research explores this new concept of dynamic music through experimentation with modular music composition — music that exists as a collection of interchangeable parts (modules) that can be reconstituted in a variety of ways. The result of this experimentation led to several interactive musical works in which various short musical modules may be played simultaneously and substituted individually to formulate an ever-changing and unique musical sum. The existence of game music environments as digital data allows new technologies such as Wwise and similar interactive game music applications to manipulate and re-sequence rendered audio files, enabling music to become an art form that can escape linearity.

Bio:

Working at the University of Newcastle, Samuel Lynch specialises in the creation of new forms of music that are dynamic, malleable, and interactive as a means of arriving at new understandings of non-linear musical forms and their impact on the creative compositional process.

Paper Session 2A — (Tuesday 10 October)

- 100 George St, Level 4. Chair: Bridget Johnson. Theme: Instruments, Interfaces and Live Electronics

Enhancing Violin Performance through Real-Time Interaction: Design and Evaluation of a Wireless Audio-Visual Interface (Anna Savery)

Abstract:

This paper presents a detailed description of the development of a wireless audio-visual interface that is integrated with a violin bow. This interface is a work-in-progress prototype resulting from a practice-based research project into optimal physical design and software mappings for the virtuoso string player. As an augmented instrument, this interface aims at extending the existing skill set of the performer. Moving away from gestural analysis approaches, we focus more directly on optimising real-time performer interaction and control for violin performance, using bow-hand fingers to manipulate the interface. A further novel aspect is the equal emphasis placed on both the audio and visual real-time processing which unlocks potential for new compositional possibilities and interdisciplinary performance situations. Frugal in its physical design, the interface is made up of two intentionally small sensors housed in a 3D-printed casing designed to prevent alterations to the violin bow. The sensors communicate wirelessly with the software system, with mappings developed within a common digital audio environment. Used within two different performance situations, this interface was evaluated from an expert performer's perspective with discussion on future design iteration possibilities.

Full paper: https://journal.computermusic.org.au/chroma/article/view/14

Bio:

Anna Savery is a violinist, composer, improviser and music technologist. Anna creates works that integrate improvised and composed music with cross-media platforms such as Max for Live, Processing and sensor-enhanced musical instruments in collaboration with multidisciplinary artists. Anna completed a Master of Fine Arts in Integrated Composition, Improvisation and Technology at the University of California, Irvine in 2016. She also holds a Bachelor's degree in Jazz Performance from the Sydney Conservatorium of Music and a Bachelor's degree in Classical Performance from the Australian Institute of Music. Anna is due to complete a PhD at University of Technology Sydney in early 2025, focusing on developing a minimally invasive wireless audio-visual violin bow interface.

Rhizomatic Improvisation (Roger Alsop)

Abstract:

This paper discusses the Rhizomatic Improvisation System for various instruments and electronics. The author has performed works in Melbourne and Prague under the pseudonym Delusion Guitari, playing guitar into a system comprising randomly timed delays and effects, with the signal from each delay and effect broadcast through a multi-speaker system. The random delay times may be from one millisecond to 20 minutes, and the feedback for each delay is infinite. This results in an improvisation system in which every improvised action is perpetually re-played into the performance environment at an unpredictable time. For the improviser this creates a situation in which each action becomes part of each subsequent and preceding action, and as these actions concatenate and coincide a sonic environment is built. The idea behind the development of this system is to create an improvisational environment in which memory and intention are linked to prediction and responsibility.

Full paper: https://journal.computermusic.org.au/chroma/article/view/15

Bio:

Roger Alsop is a multidisciplinary artist who focuses primarily on the creation of collaborative and improvised artworks. He works in theatre, galleries, and music performance and his work has been presented internationally. His work focuses on relationships between text, sound, and visual imagery, and how these may be generated and interpreted through collaborative, improvised, and computer-based processes. He has lectured and presented works at Nicola Sala Conservatorium and the Greenwich, Edinburgh, Zagreb, and Belgrade Universities, and presented artworks and writing at the International Computer Music, Korean Electro-Acoustic Music Society, and Australasian Computer Music Conferences, ISEA, CSIRO, Prague Quadrennial, World Stage Design, and the Melbourne Festival. He supervises research students and teaches at Melbourne University and Box Hill Institute.

Songs in Isolation: Social Commentary and Contagion, COVID-19, and Creative Practice in Electronic Music (Sophie Rose)

Abstract:

Songs in Isolation is a series of nine mini-works for online multimedia distribution. The works are a time capsule of the first 18 months of the COVID-19 pandemic. Each piece was informed by Melbourne, Australia's global and locational socio-political contexts. Composition themes include frustration (at public messaging, government incompetence, panic buying, lobbying by large businesses at the detriment of the rest of society, and spreading of disinformation), socially contagious behaviours (home hairdressing, meditation, dieting), and personal reactions to stress (brain fog, dissociation, pressure to generate creative content). This research is practice-based performative autoethnography. The pieces and social context move together, buffeted by the movements of a virus and the success or failure of each country concerning travel movements and location-specific behaviours of the public. Various composition tools and techniques were used to suit each work, including gestural interfaces (Wave, MiMu, self-built datagloves – GLVD), a consumer EEG interface (Muse 2), various DAWs and field recording devices, and voice.

Full paper: https://journal.computermusic.org.au/chroma/article/view/13

Bio:

Sophie Rose is a multi-instrumentalist, singer, extended-technique enthusiast, composer, improvisor, researcher, multi-media artist, and maker. She is currently undertaking a PhD at The University of Melbourne in Interactive Composition. Her work explores the relationship between creative practice, interactive technology, trauma, feminism, and embodiment. Sophie's masters examined the role of physical versus virtual environments and their effect on composition for extended vocal techniques. In performance works, Rose mixes technology and technical proficiency to explore the nexus between human potential and the affordances of machines. Her work has been featured in NIME and Riffs Journal.

Paper Session 2B — (Tuesday 10 October)

- 100 George St, Level 4. Chair: Charles Martin. Theme: Time, Phase and Performance Media Arts

Musical Congruency and Visual Aesthetics: an Auto-ethnography in VJ Performance (Mark Oliveiro)

Abstract:

The art of the VJ (visual jockey) serves as a representation of musical structure within the international electronic music scene. When a VJ curates musical performances in these communities, they have the opportunity to visually embody the essence of the music, especially when there is a preceding visual corollary. Various sub-genres of electronic music, including Electronic Dance Music, often associate with specific visual art movements or aesthetics derived from film or animation, which contribute to the overall musical experience. This study seeks to examine the prevalence and consistency of these artistic ethnographies as exclusive dualities, or whether the visual aesthetics of these musical movements can be rearranged and remixed in a manner similar

to the music itself. Drawing on unique perspectives and data from the recent electronic music club scene in Sydney, Australia (2022–2023), this research focuses on the diversity and dynamics of aesthetics in genre-specific musical events and their associated visual spectacle, with a particular emphasis on VJ art and performance: How musically congruent is contemporary VJ art, and to what extent should vision and sound be interconnected within this context?

Bio:

Dr Mark Oliveiro is a graduate and distinguished alumnus of the Sydney Conservatorium, the University of North Texas and the School of Music of Indiana University. As well as working as an academic lecturer at AIM, Mark is a composer with diverse extra-musical interests including history, mythology and ancient literature. At present, he has an interest in exploring the notion of reconstructing lost or imagined performance traditions. Dance and movement, theatrics and technology, language and social media are just some of the non-musical solutions apt for his ever-expanding, contemporary composer tool belt. Mark's music has been represented at national and international conferences and festivals including the World Saxophone Congress, the International Horn Society, the International Computer Music Association, the Society of Electroacoustic Music, Electronic Music Midwest and Vivid Music.

Playing With Time (Neil Steward)

Abstract:

Time or temporality is generally perceived as something absolute, and with the introduction of accessible and common computing power, time has become even more structured and managed — yet the role of temporality and its relationship with music can be seen to be something altogether different, or as Berger (2014) says "music hijacks our relationship with everyday time". As part of the compositional process of some works, various forms and techniques are used and, as a result, a listener's perceptions of time may be challenged. Among the forms used are linear/non-linear time, vertical time, moment time and polychrony, and techniques such as discontinuity, avoidance of western tonal harmony, lack of clear rhythmic material and repetition. The work *Water*, created as part of my PhD project, demonstrates the use of a number of these techniques. It uses a part played by multiple instruments, each playing the part at a different tempo but all beginning at the same moment. As the work progresses, various instruments play the part transposed a fifth or an octave, inverted or reversed, yet remaining at their initial tempo.

Bio:

Neil Steward is a Creative Practice PhD student at University of New England working in the field of electroacoustic music with the aim of composing pieces that investigate time and timelessness. When not composing in the field of electroacoustic music he works in the genres of folk, pop, and rock both in the studio and in live situations. He lives in Brisbane with his wife and his dog.

Computation and Musical Applications of Discrete and Continuous Rhythmic Phase-shifting Techniques (Yoonjae Choi)

Abstract:

Based on the rhythm phase-shifting traditionally used in Western music, this paper describes how to implement it in a computational way and how to shift the rhythm phase beyond the scope of the conventional sphere. It is mainly divided into two approaches: discrete phase-shifting, which can be implemented through the shifting of the sample delay, and continuous phase-shifting through the adjustment of the playback speed (itself divided into two techniques depending on whether the playback speed is fixed or changed). It also deals with the mixing of these various techniques.

For each technique, we discuss the process and its mathematical formula that must be applied to align phase-interval in rhythm units of a given BPM.

Full paper: https://journal.computermusic.org.au/chroma/article/view/18

Bio:

Yoonjae Choi studied composition and electronic music with Richard Dudas at Hanyang University. He is currently enrolled in the Computer Music Composition program at Indiana University Jacobs School of Music. As a composer, he is influenced by various genres such as contemporary music, minimal music, and pop music. He is researching ways to artistically utilise computers and digital technology based on his musical ideas.

Paper Session 2C — (Tuesday 10 October)

- 100 George St, Level 4. Chair: Jordan Lacey. Theme: Science, Systems and Improvisation

Inspired by Science: How Interdisciplinary Forces Can Transform Electronic Music (Alexis M Weaver & Geosmin Turpin) — Artist Talk

Abstract:

This joint Artist Talk discusses the creative process of developing *Soft Matter, Hypersensitive Instruments* (SMHI), a collaborative electronic work commissioned by the Venice Biennale for October 2023 premiere, featuring poetry by Dr Geosmin Turpin and electronic music by Alexis Weaver. SMHI posed a unique developmental challenge; inspired by the concept of Atomic Force Microscopy (AFM), the work comprises the output of two artists working iteratively and in parallel. While Dr Turpin is a post-doctoral chemist and poet, Weaver is an electronic music academic. Both artists' contributions were inspired by the other in real time. Furthermore, the SMHI project revealed several insights into interdisciplinary collaboration, and the effect of external resources and stakeholder input into the creative process. SMHI represents a delicate balance between a more didactic communication of science and artistic freedom. Weaver and Turpin discuss the artists' response to the initial stimulus and to each other's output, the evolving intentions of the work, the 'ideal' pressures and parameters for 'successful' artmaking, and the takeaways of this year-long shared practice. The talk is accompanied by excerpts of the final work in stereo format.

Bio:

Alexis M Weaver (she/her) is a composer, sound artist and educator based in Sydney, Australia. A passionate composer of electroacoustic music, Alexis has also composed soundtracks for animation, short film, theatre, dance, and a series of podcasts for Australian companies. Her electroacoustic works have been broadcast in Australia, the US, the UK, and Europe. In 2018, Alexis was awarded the National Council of Women's Australia Day Prize for her research on the visibility and practice of female electroacoustic composers. She completed a Master of Music at the Sydney Conservatorium, where she now lectures, and has commenced her PhD examining the role of music and sound design in science communication.

Dr Geosmin Turpin is an Australian poet-chemist named after the molecule responsible for the smell of rain. She completed her PhD in physical chemistry in 2022, studying the boundary issues of nanoparticles at Monash University before continuing her exploration of nanoscale systems at the University of Sydney. She enjoys taking the purposeful language of science and exploding it into poetry. Geosmin is joyfully transgender and proudly neurodivergent. Her poetry has been published in *vox populi* (Bowen Street Press) and *In Flux* (Thousand Threads Press). More of her work is available at https://uncleftwords.com/.

Building Musical Systems: An Approach Using Real-Time Music Information Retrieval Tools (Austin A Franklin)

Abstract:

This research presents a new approach to computer automation, termed Reflexive Automation, through the implementation of novel real-time music information retrieval algorithms developed for this project. It introduces the development of the PnP.Maxtools library, a 'plug and play' set of filters, subjective timbral descriptors, audio effects, and other objects developed in Cycling '74 Max, designed as a framework for building musical systems for music using live electronics without the use of external controllers or hardware. The library takes incoming audio from a microphone, analyses it, and uses the analysis to control an audio effect on the incoming signal in real time. This approach to automation and library together represents the first defined method for performance and composition practice using music information retrieval algorithms.

Full paper: https://journal.computermusic.org.au/chroma/article/view/16

Bio:

Austin A Franklin is an internationally recognised composer and sound artist based in Baton Rouge, LA. His interests include music involving process, such as algorithmic composition and music incorporating music information retrieval and machine learning technologies. His album *Four Idols* (2020) has been described as "an elegant, artistic statement that demonstrates the flexible possibilities of electronic music". He has pieces for percussion published through C-Alan Publications and his music has been performed throughout North America, South America, Europe, Africa, and Asia. In 2022, he made his Carnegie Hall debut. He is the recipient of numerous awards and commissions, including the American Prize for Composition, and has presented research at the Web Audio Conference (WAC) and NIME. For more, visit austinfranklinmusic.com.

The Evolution of Electronica Bass: The Bass as an Interface for Electronic Music (Mansell Laidler)

Abstract:

This paper outlines the practice-based research undertaken to explore the ways in which the electric bass can be used as an interface for the creation of live, improvised electronic music. Specifically, it examines how an experienced bass player can build upon existing techniques and expand their performance practice in a solo, improvised context. While the electric bass is traditionally seen as a supporting instrument in contemporary ensembles, its solo repertoire is relatively limited. This paper outlines the limitations and challenges associated with performing solo bass works and proposes strategies and processes for developing new responsive performance frameworks. It illuminates the development of the project from one about solo bass to the use of the bass as an interface for electronic music. The paper provides an overview of the progression of the artist's practice, starting with early experiments involving looping technologies, through to current techniques with Ableton Live and the Line 6 Helix LT. Drawing inspiration from notable virtuosic bass players like Michael Manring, Janek Gwizdala, and Cody Wright, the artist explores innovative techniques and tools to expand the sonic possibilities of the electric bass.

Bio:

Mansell Laidler is a multi-talented artist from Canberra, Australia — a bass player, electronic musician, researcher, and instrumental teacher. For the past decade, Mansell has toured extensively across Australia, gaining a wealth of experience and an intimate understanding of live music. Now based in Canberra, he has embarked on a diverse array of creative projects, fearlessly exploring new sounds and artistic expressions, most notably his current research project.

As well as a gifted performer, Mansell is a dedicated instrumental teacher, combining technical mastery with a deep understanding of musical theory.

Paper Session 2D — (Tuesday 10 October)

- 100 George St, Level 4. Chair: Neil Steward. Theme: Electronica, Ritual and Afterlife

From Ritual Music to Electronica: The Transformation of Traditional Santeria into Electronic Neo-Santeria (Vincent Sebastian Labra)

Abstract:

Santeria is a Cuban religion derived from West African Yoruba musical traditions. Santeria practitioners believe that music is a practical technology that alters consciousness. This research explores the transformation of traditional Santeria acoustic music into contemporary forms of electronic neo-Santeria by exploring the emergence of a neo-Santeria style in international music and its development from traditional Santeria religious rituals. The research used a mixed-method approach combining musical analysis, auto-ethnography, and practice-led creative research to explore the objective and subjective nature of music within its traditional and secular performance contexts. The outputs included music composition, live performances, and the production of music EPs, album cover artworks, a music video, and visual records of live performances. The study found that the emergence of neo-Santeria in electronic music is a multi-sensory transformation, incorporating musical, visual, conceptual, and performative elements. The research concludes that the innovations in neo-Santeria develop from a set of underlying ritual principles that reflect cultural objectives for altering consciousness.

Bio:

Vincent Sebastian is a PhD researcher, percussionist, music producer, and DJ based in Sydney, Australia. He specialises in Latin American cultural music and electronica and has an extensive career performing and creating music in Sydney and abroad. His accolades include an ARIA nomination and performances supporting artists such as Santana, Seun Kuti, Ricky Martin, Los Amigos Invisibles, Los Van Van, and Earth Wind and Fire, as well as performances at Bluesfest, Woodford Folk Festival, and international festivals in Malaysia, Colombia, Korea and Japan. His PhD research explores an emerging style of electronic music that has developed from Cuban religious rituals, realised through his electronic music group Oyobi. Vincent also manages 'The Nest', an inner-city creative hub dedicated to music production, recording, and composition.

Drip — Workshop Demo (Finn Mitchell-Anyon / Finneganeganegan)

Abstract:

Drip is a VST plugin which provides a new interface for generative music composition. The user is able to place water taps at the top of the screen. These taps emit water droplets at regular musical intervals. The user can also place bars which, when struck by a water droplet, send out a MIDI note. The pitch of the note is dependent on the length of the bar, which the user is able to control. The velocity of the note is dependent on the force with which the bar is struck. This interface allows the user to organically create musical sequences by arranging elements on-screen. Since the plugin outputs MIDI, notes can be sent to any VST instrument or hardware synthesizer.

Bio:

Finn is a music producer, sound designer, game composer, and programmer. Graduating from Victoria University of Wellington in 2019, his audiovisual work explores new means of musical expression.

AFTERLIFE (ONLY SLIME — Tobi Pfeil & Claudia Cox) — Artist Talk

Abstract:

AFTERLIFE is an electronic computer-game opera crafted by the interdisciplinary arts and music duo ONLY SLIME, composed of Tobi Pfeil, a composer and media artist, and Claudia Cox, an artist and musician. The game opera challenges traditional notions of life after death and the pursuit of deeper meaning by merging elements from Greek mythology, contemporary internet culture, AI-generated content, and computer-game storytelling techniques. Presented in a 3D-game-inspired operatic format, AFTERLIFE melds entertainment, thought-provocation, and innovation in music, technology, storytelling, and live performance. Using real-time motion tracking and live vocalisation, alongside electronic elements on stage, the two performers guide their avatars within a game engine throughout their existential quest for deeper meaning, including a descent into the realm of 3D death and non-existence.

Bio:

ONLY SLIME is a multidisciplinary music and digital arts duo by Claudia Cox (aka Peri Winkle) and Tobi Pfeil. Their work includes interdisciplinary art in the intersection between music, game design, AI-assisted art, performance, visual works, installations and 3D animation, with a strong emphasis on collaboration and interdisciplinarity. Only Slime was founded in 2022 and has since produced their debut computer-game opera *AFTERLIFE*, scheduled for official premiere at Black Box Theatre in Oslo 2024.

Paper Session 3A — (Wednesday 11 October, 9–10.10am)

- 100 George St, Level 4. Chair: Blake Johnson. Theme: Live Electronics, Education and Inclusion

The Haptic Fader Audio Meter: Inclusion and support for blind and low-vision music and audio creators in the digital era (James Hurley)

Abstract:

This paper presents the initial analyses and findings from interviews and task observations conducted with blind and low-vision participants who use Digital Audio Workstations (DAWs) in their creative work. This data informed the direction of participatory design sessions that developed prototypes in the haptic domain to address the accessibility and usability need-gaps identified to date in our research. Current assistive technologies for this population, which have been adapted from standard computer access technologies and utilise the sole interaction modality of computer speech feedback, can be problematic when the user's auditory perceptual channel is already fully engaged with creative content. We present a prototype vibrotactile haptic feedback device that aims to address one of these accessibility barriers — the monitoring of audio meter levels while recording, editing, and mixing audio. The ability to monitor audio signal levels is crucial in all music and audio production fields, and the accessibility and usability of speech feedback assistive technology for this purpose has been identified as a key barrier for our research population.

Bio:

James Hurley worked for over 30 years in music composition and sound design in Australia and internationally for film, television, live performance, exhibitions and installation. James is conducting research with blind and visually impaired users in the co-design of haptic (touch-based) feedback to enable better accessibility and usability in the graphically dense and cluttered software interfaces used in contemporary digital music and sound production.

Teaching 14 Laptop Ensembles (Charles Patrick Martin)

Abstract:

At ACMC 2019, my colleagues and I asked whether our laptop ensemble course-design experiments could scale to 50 students (Swift et al., 2019). This year, we had 60 students and I'm going to tell you all about it. Since 2018 work has been ongoing at ANU to teach computer music across computing and music curriculums, posing experiments with course designs influenced by computer science teaching. This paper examines the journey through laptop ensemble curricula from 2018 to now, with a focus on scaling to 14 laptop ensembles in one semester and the challenges of teaching primarily computer science students. I introduce our course design (Martin, 2023) where standard computer music coding is blended with interface design, networking, and collaborative performance, and draw particular focus on our workshop designs, influenced by Murphy's (2019) "Beat Cypher" pedagogy. I demonstrate course automation tools adopted from large computing courses, and consider the role of music within computer science education. Where our previous paper asked how a laptop ensemble class might scale, this paper asks what these students might get out of the experience.

Bio:

Charles Martin is a computer scientist specialising in music technology, creative AI and human-computer interaction at The Australian National University, Canberra. Charles develops musical apps such as MicroJam and PhaseRings, researches creative AI, and performs music with Ensemble Metatone and Andromeda is Coming. At the ANU, Charles teaches creative computing and leads research into intelligent musical instruments. Prior to his present appointment as a Senior Lecturer in computer science at the ANU, Charles was a postdoctoral research fellow at the University of Oslo's Robotics and Intelligent Systems research group (2016–2019).

Ode and Melody: A Live Music and Story Performance (Lionel Hobden)

Abstract:

Ode and Melody is a live musical performance targeting stage 2 primary school students, which includes multi-modal learning facilitated by music, narrative, art and stage play. The aim of the project is to inspire younger generations to take up an instrument, engage in live music performance and to deliver positive messages about issues they may be facing at school — dealing with your feelings, feelings of failure or anxiety, learning that it is okay to make mistakes, and bullying. The performance is a 20–30 minute set of storytelling through song and art, using traditional instruments such as guitars and percussion alongside digital instruments manipulated via an electric guitar with a MIDI pickup. The performers are bards within the world of Ode and Melody. In order to link the performance to children's education, the project uses the NSW primary school syllabus and the Australian Student Wellbeing Framework as guides.

Bio:

Lionel Hobden is a music writer and stage coordinator currently studying at UNE. He is a guitarist/musician of over 25 years with a keen interest in mixed-media performances. He has played in original bands from 1999 until COVID stopped performances, and now wants to teach music for a living and inspire others to play. The Ode and Melody project is a new venture taking Lionel out of his comfort zone.

Paper Session 3B — (Wednesday 11 October, 9–10.10am)

- 100 George St, Level 4. Chair: Toby Gifford. Theme: AI and Digital Spaces

I Sing the Body Electric: Human-Machine Interactions at the Frontier of Creativity and Artificial Intelligence in the Making of Music (John Hawkins)

Abstract:

The symbiotic relationship between human creators and so-called computational creativity systems has grown exponentially over the last few decades. Where once machines were seen as merely tools, they are now seen by many artists as collaborators and co-producers. Recent developments in generative AI have primarily focused on apps for textual and image production, such as ChatGPT and DALL-E 2, but there is also a growing array of digital AI tools to help a human creator produce music. In this paper, I consider a small sample: Google's MusicLM; Lango Rhythm; Bot Dylan; AIs that finish where classical composers left off (Schubert's "Unfinished" Symphony and Tchaikovsky's abandoned 7th Symphony); and AIs that create new symphonies (Beethoven's 10th) or an endless string of new works by Bach. These raise important new questions about the creative process, the ethics of ownership of AI-generated compositions, and how the symbiosis between humans and AI will change the way we think about creativity. I argue that despite impressive developments, computational creativity AIs still fall short of the "Turing Test" of producing autonomous human-like creation.

Bio:

John Hawkins is a doctoral student in the philosophy department at the University of New England, researching the future of human consciousness in the age of AI. He has longed to compose a symphony but felt hopeless without the requisite musical repertoire of skills; recent AI applications have now made it possible for him to pursue this quest. In addition, Hawkins is a poet and freelance journalist.

DuoTube: Inspiring Experimental Creativity in Common Digital Spaces (Ralph Lewis & Robin Meiksins)

Abstract:

In *DuoTube: Inspiring Experimental Creativity in Common Digital Spaces* we discuss the branching inspiration throughout a series of creators that came about from creating the piece *DuoTube*. Originally composed specifically with flutist Robin Meiksins' YouTube channel's pre-pandemic aims at creating digital performance spaces in mind, DuoTube reuses YouTube's functions to create an interactive computer part to play along with Meiksins' performance. Through testing and performances, it became clearer that it offered a different kind of accessibility than most works — one that could support solo or group performances, whether in the concert hall or at home. The reuse of YouTube's functionality, portability, and scalability has since inspired a number of collaborators. Throughout, we reflect on how this series of pieces and its experimental situations in familiar spaces inspire different kinds of electronic-music-related stakeholders.

Bio:

Ralph Lewis is a composer whose works seek meeting points between sonorous music and arresting noise, alternative tunings and timbre, and the roles of performer and audience. Lewis's music has been presented at festivals and conferences including SEAMUS, TENOR, SCI, Boston Microtonal Society, College Music Society, Electronic Music Midwest, MOXsonic, and more. Lewis currently serves as SEAMUS's first Member-at-Large for Outreach.

Robin Meiksins is a freelance contemporary flutist focused on collaboration with living composers. While Chicago-based, she uses the Internet and online media to support and create collaboration. In 2018, Robin launched the 52 Weeks of Flute Project. She has premiered over 100 works by living composers and has performed at SPLICE Institute, SEAMUS, Oh My Ears New Music Festival, and Frequency Festival.

Warped Reflections: Working with AI to create new, composer-specific sound worlds (Alexis Weaver & Rowan Savage) — Artist Talk

Abstract:

This Artist Talk covers the technological and creative processes behind *Sonic Mutations*, a collaboration between a new AI model and electronic sound artists Rowan Savage (salllvage) and Alexis Weaver. First commissioned by the Sydney Opera House for the 2023 Outlines Series, Sonic Mutations involved the creation of a new AI tool based on 'Riffusion,' a neural network which creates sound from AI-generated spectrograms. Originally conceived by Kopi Su studios and developed with creative technologist Adrian Schmidt, the result is a live performance and panel event. From conception to execution, the Sonic Mutations project challenged the artists to discover how the use of bespoke AI could enhance and grow their existing creative practice. The new AI tool became a composer-specific tool trained on the previous outputs and chosen prompts of Rowan and Alexis. The outputs, while identifiably true to the composers' established artistic voices, created highly original novel outputs — showing that such AI-powered sound generation tools do not necessarily threaten, but can expand a living composer's practice.

Bio:

Alexis Weaver (she/her) is a composer, sound artist and educator based in Sydney, Australia (see Paper Session 2C).

Rowan Savage (salllvage) is a proud Kombumerri man, living on Wangal Land. He is an experimental producer and DJ working at the intersection of queer club culture and connection with Country. His work mutates on-Country field recordings into electronica, and inhabits and bridges the tensions between abstraction and emotion, the wild bush and the dancefloor, the personal and the social, authenticity and reconstruction. Rowan has a background in music production, working closely with Ableton combined with a longstanding field recording practice.

Pieces and Paper Sessions

Piece and Paper 1 — Approaches to Spatial Audio for Audiences of All Types of Hearing (Jesse Austin-Stewart)

- Monday 9 October, 11.30am–12pm — 100 George St, Level 4. Chair: Roger Alsop

Abstract:

Spatial Audio is a field of music that relies on audiences having equal and full-range amplitudinal and spectral hearing in both ears in order to accurately perceive spatialisation. In acknowledging this issue, the author has engaged with the hard-of-hearing and d/Deaf music community and created various compositional strategies that can be used to aid spatial audio composition for audiences of more broad types of hearing. The paper outlines previous work within the field, summarises interviews with the d/Deaf and hard-of-hearing community, discusses the new strategies developed and new works developed based on those strategies, and the responses from the community to the newly developed works. The works *Spatial Vibrations* and *Music for*

PlayStation were exhibited alongside the paper as installations demonstrating some of the spatial strategies developed.

Bio:

Jesse Austin-Stewart (he/him) is a sound artist based in Te Whanganui-a-Tara Wellington, Aotearoa New Zealand. He is a composer, sound artist, producer, and academic and is currently a lecturer at Massey University in the School of Music and Creative Media Production Te Rewa o Puanga. As a sound artist, Jesse creates works that explore ways to make the field of spatial audio more inclusive, often exploring the intersection of spatial audio and disability and hearing, while also acknowledging barriers of finance and education. His work has been exhibited in Aotearoa New Zealand, Japan, Australia, Czech Republic, Greece, Sweden, and France. As a person with a disability, accessibility is core to Jesse's work and his artistic curation.

Piece and Paper 2 — A new ecosystem for microtonal computer music exploration and composition (Warren Burt)

- Monday 9 October, 2–3pm — 100 George St, Level 4. Chair: Jesse Austin-Stewart

Abstract:

Over the past year or so, a number of new programs have emerged which make the composing and exploring of microtonal music much easier and more efficient. In the interaction of these programs, a new eco-system to support microtonal music has come together, greatly expanding the possibilities for microtonal performance, especially in real-time. This paper looks at several of these programs and how they interact to create a new, more powerful composing environment. The principal programs are MTS-ESP by Oddsound, Wilsonic by Marcus Hobbs, and VCV Rack Pro by Andrew Belt and associates. As more developers have come on board supporting the MTS-ESP standard, the number of microtonally capable instruments has expanded greatly. Also, "framework" programs such as Ableton Live, Bitwig Studio, and Reaper make the implementation of this ecosystem quite easy.

Full paper: https://journal.computermusic.org.au/chroma/article/view/17

Program Note:

Using the Pentadic Diamond page of the Wilsonic program, with three instruments (Arturia DX7, Pianoteq, and Surge XT), a 3-voice random-walk canon made with Algorithmic Arts MusicWonk is performed. The first minute uses the 21-note master set, then 10 sections of 30 seconds each go through the 10 five-note subsets, then a final minute of all 21 notes again.

Bio:

Warren Burt is an independent composer based in Ararat in Western Victoria. He has performed his music all over the world in a career spanning more than 50 years. He is currently involved in beta testing a number of microtonal music programs.

Piece and Paper 3 — Zen of Aggression: Composing the Transformational Hybrid (Berk Yagli)

- Monday 9 October, 3.30–4pm — 100 George St, Level 4

Abstract:

The presentation focuses on musical hybridity and critical methods developed for the hybridization of electroacoustic and metal music (part of the presenter's PhD research). These methods build a blueprint for hybrid composers — not only specific to metal and electroacoustic but for

other genres that combine spectromorphology-focused music with harmony and rhythm-based music. The talk focuses on the piece *Zen of Aggression* and discusses the fruitful and problematic methods when approaching to compose a transformational hybrid (a hybrid where both genres affect each other at the deepest level, so the resulting hybrid does not sound like either genre). Different excerpts from the piece are played when discussing the techniques and methodologies. The talk also briefly discusses different types of hybridity (transformational hybrid, eclecticism, polystylism), genre and its notions, and the notion of fluidity when approaching hybridity in music.

Program Note:

In a world where saturation becomes the norm, little to no values are left without being commoditized, and opinions become one-dimensional, the people living in society evolve to become numb and alienated from overstimulation. *Zen of Aggression* deals with these emerging conditions to produce an electroacoustic/metal hybrid piece as aggressively as possible.

Bio:

Berk Yagli (born 1999) is a Cypriot guitarist, composer, and producer. His mission with his music has been to talk about social, political, and philosophical matters to invite listeners into reflecting on the topics. He has been active in the UK for the past six years due to his education in Music and Sound Technology (University of Portsmouth), Masters in Composition (University of Sheffield), and currently at the University of the Arts London working under Adam Stanovic for his PhD topic on hybridity between metal and electroacoustic music. He composed and released a cinematic/epic social-commentary progressive metal album *Symphony of Humanity* in 2021. His works have been presented in Europe, Asia, North and South America, and he won the 2022 18th WOCMAT Phil Winsor International Youth Computer Music Competition Award.

Piece and Paper 4 — Sonic Crucible: a new materialist exploration of the quartz singing bowl (Jordan Lacey & Toby Gifford)

- Tuesday 10 October — UNE Installations, 28 George St, Ground Floor

Abstract:

This piece & paper applies a new-materialist examination of New Age claims regarding quartz singing bowls. These claims include a belief that the bowls' resonant properties, when tuned to the Western Major scale, activate chakra points. We discuss several issues surfaced by this claim: it connects two distinct cultural practices (the Western tonal system and embodied chakra systems); it appropriates ancient Buddhist rituals; and the actual history of the quartz singing bowl is ignored. The true history of the quartz bowl is its role as a crucible, with which the Czochralski Method was applied to fabricate silicon microchips for the 1980s computer industry. The paper critiques New Age theoretical conflations, and through a conference-based performance presents the quartz singing bowl as an installation entitled the *Sonic Crucible*. The paper concludes by arguing that the New Age is the natural spiritual expression of Capitalism, and uses the installation-performance to argue that when materiality is met body to body, affective experiences can occur without the need for economic, historical and cultural appropriation.

Program Note:

The installation is a quartz singing bowl set upon a stand and plate spun by a motor. A projecting stick presses a silicon rod onto the bowl, causing it to resonate upon rotation. The bowl holds water, with a suspended hydrophone playing resonant sounds through two speakers at the installation base. The rate of rotation and the processed sounds are controlled using a real-time app built with Pure Data. The Sonic Crucible was installed for the duration of the conference and used as an instrument for a one-off performance with accompaniment and sonic activation by a contrabass clarinet, chosen due to their common property of low-frequency drones.

Bio:

Dr Jordan Lacey is a sonic thinker, creative practitioner and transdisciplinary researcher who specialises in sonic theory, soundscape design and the creation of public sound art installations. He is based in the School of Design at RMIT University. He has written two books with Bloomsbury — *Sonic Rupture: a practice-led approach to urban soundscape design* (2016) and *Urban Roar: a psychophysical approach to the design of affective environments* (2022) — and has been the recipient of two competitive research grants including the Australian Research Council DECRA Award (2019–2022). He is Associate Editor and lead Book Review editor for the *Journal of Sonic Studies*. A comprehensive list of his works can be found at jordan-lacey.com.

Toby Gifford is a sound technologist with a diverse array of research interests under the banner of 'sonic environments'. As a sound-artist and musician he works with multichannel, binaural and ambisonic sound for virtual and augmented reality, immersive installation, and live performance. His practice utilises generative algorithms alongside field recording, with application in bioacoustics, aural architecture and computational creativity. He is on the boards of both the Australian Forum for Acoustic Ecology and the Australian Ecoacoustics Society, and was general chair of the 2016 New Interfaces for Musical Expression international conference. He holds a PhD in computer music & AI from QUT, a Masters in pure mathematics from Washington University in St. Louis, and first-class honours in mathematics & physics from ANU.

Installations

PHIVE Installations

- PHIVE, 5 Parramatta Square, Parramatta

AI Robot — Keirzo (Richard Savery)

Program Note:

Keirzo is a new robot for rapping and drumming, developed by Richard Savery as part of his work as a Macquarie University Research Fellow exploring AI, creativity, music and robotics. Keirzo can rap for any length of time with humans of any musical skill level, from professional rappers to non-musicians. Responding to input phrases, rap, or single words, Keirzo produces an original rap in real-time along with a groove. Another feature generates a one-minute composition — a bass line, chords, harmony lines and lyrics — from three words input by a human, then plays and raps the new composition. More details are at the robot's website.

Bio:

Dr Savery is a developer of artificial intelligence and robotics, using music and creativity as a medium (see Keynote 2). https://richardsavery.com/

Robot Hang Drum Installation (Jon Drummond)

Bio:

Jon Drummond is Associate Professor of Music in the School of Humanities, Creative Industries and Social Science. Jon's creative work explores interactive electroacoustics, robotics, sonification of natural phenomena, acoustic ecology, and real-time interactive performance systems for acoustic instruments. His works have been presented at many festivals and conferences, including the Adelaide Festival, ISEA, ICMC, NIME, and the World Forum for Acoustic Ecology. His research interests include human-computer interaction design, new interfaces for musical expression, gesture analysis, improvisation, sound spatialisation, data sonification and artificial intelligence.

Stochastic Determinism (April Guest)

Program Note:

Stochastic Determinism is an interactive sound installation where people can manipulate a continuously playing EDM track using their voice. Microphone input from public participants runs through generative algorithms to change elements of the music in real time. The installation explores the ambiguous, mutable relationship between body and instrument by magnifying participants' inherent human imperfections and obfuscating the mechanism between input and output. The algorithm generating the music is deterministic, however some aspects feel impossible to control due to human error while others do not, creating a feeling of "quasi-randomness" or "indirect control" where users sense their own imperfection. Stochastic Determinism is a reflection on how human and machine entwine and a metaphor for the imperfect control we have over our everyday lives.

Bio:

April Guest is a Naarm-based vocalist, beatboxer and composer. Her work focuses on pushing the sonic capabilities of the human voice both biologically and technologically. Her live-looping, ambient performances feature extended vocal and beatbox techniques to create deep, expansive drones and intricate percussive layers. She is also interested in exploring microtonality to showcase the under-utilised expressive qualities of pitch and tuning. She also performs and composes under the alias Byte Crusher in an EDM context.

UNE Installations

- 28 George St / 100 George St

Immersion (Lisa Griffiths)

Program Note:

Immersion is an audio-visual artwork designed specifically for festivals. It combines 3D projection mapping with dance choreography and is accompanied by original and diverse world music. This artwork invites festival-goers to embark on an immersive journey, celebrating festival culture itself while embracing the mysterious ambience that surrounds it. Audience members walking through the stalls pass a marquee tent and, on stopping, realise that what they see inside is an illusion mixed with reality — the mystique.

Bio:

Lisa Griffiths is completing an Honours degree at UNE under the supervision of Associate Professor Donna Hewitt (see Paper Session 1A).

Fractures and Faults (Vicki Hallett)

Program Note:

Fractures and Faults is a long-form piece using sounds from a natural artesian spring within the Great Artesian Basin aquifers, Australia. This composition uses field recordings from a series of hydrophones placed within the natural spring where the groundwater seeks fractures, faults and folds. Hear the underground turmoil as the pressurized water forces its way through the Fracture Zone, rises to the surface, escapes the discharge vent and flows towards the wetland. The Great Artesian Basin is a vital groundwater resource covering four states of Australia over 1.7 million square kilometres, and has cultural significance for Aboriginal people for their Creation stories and as a vital water source.

Bio:

Vicki Hallett is a musician, acoustic ecologist, sound artist and composer. Hallett's focus is to record, compose with, and feature ever-diminishing habitats and species. She seeks spaces with limited anthropogenic sounds and infrasound interactions, highlighting unique sounds from the micro to the macro. She has composed, produced and performed in live concerts, international conferences, solo recordings and videos ranging from chamber music to exploratory sound art. She travels the world recording nature's sounds and performing in acoustically-interesting environments such as Mabolel Rock (South Africa) with a pod of hippopotami and the Amazon jungle.

Time xCells (Charlie Morton, Stephen Bainbridge & Genevieve Van Staden — Massey University)

Program Note:

It is virtually impossible to complete a full working or leisurely day without encountering technology designed to grab and maintain your full attention. *Time xCells* attempts to display a contemporary visual representation of the toll algorithmically targeted media consumption takes on the human psyche and body. The visual element is a digital video featuring manipulated prerecorded material and overlaid graphical animations. Accompanying the video is a synchronised music piece drawing from the soundscapes of downtempo electronica, psychedelic trance and spoken word. Featuring lyrics in a diary-like form, they detail the character's experience from a first-person perspective. As the piece progresses, their voice distorts and chops before blending into the electronic production of the beats, succumbing to their media addiction, never being heard from again.

Bio:

Stephen Bainbridge is from Wellington, New Zealand, and predominantly makes experimental music with influences from trance, studying for a Bachelor of Commercial Music majoring in Music Technology at Massey University. Charlie Morton was born and raised in Auckland before moving to Wellington to study Music Technology at Massey; he has worked in video and audio and released his debut EP under the alias "charmort", drawing on IDM, ambient and drill 'n' bass. Genevieve Van Staden moved to New Zealand in 2016 and is studying a Bachelor of Commercial Music majoring in Music Performance at Massey University, Wellington.

intro/exo (Blake Johnston — Massey University)

Program Note:

intro/exo is a composition for a bespoke headset that allows the audience to experience extraordinary spatial and sonic phenomena. The headset consists of two loudspeakers, which sit on the ears like normal headphones, and two surface transducers that sit on the cheekbones of the wearer. These transducers vibrate the cheekbones and skull, allowing the experience of sounds seemingly emanating from within the head. The work investigates the creative and experiential affordances of the bespoke headset by placing the listener's head as the site of exploration, creating sounds that move around and inside the listener's body. The work employs binaural techniques to create hyper-realistic sonic environments, contrasted with the intimacy of the surface transducers. It is the second in a series of works that explore the creative potentials of these interfaces.

Bio:

Blake Johnston is a sound artist, technologist, academic, and composer from Aotearoa, New Zealand. His research practice sits at the intersection of experience design and emerging forms of technology, synthesising these fields to explore the perception of the audience. His work adopts new forms of technology to create environments and experiences across multiple disciplines including live performance, kinetic sculpture, musical mechatronics, wearable technology, and installation art. He is an active educator teaching composition, interface and interaction, human-computer interface, music technology, software and hardware development, and mechatronics.

Life (Enrico Dorigatti — University of Portsmouth)

Program Note:

Life is a generative audiovisual composition based on an adapted version of the famous Life algorithm developed by mathematician J. H. Conway. The adaptation includes a three-dimensional environment in place of the original two-dimensional one and other mechanisms (such as ageing and illness) which increase the complexity of the algorithm concerning whether a cell should be born, die, or survive the next epoch. Whilst all these processes occur in the background, hidden from the audience, they are showcased in the visuals and sounds. Both media primarily represent the status of the algorithm or parts of it, crossed with other parameters or manipulated through filters, favouring an artistic representation of the data over a faithful one. *Life* is an example of pure algorithmic audiovisual composition; the first version was written in 2017.

Bio:

Enrico Dorigatti is an experimental sound artist and creative technologist. He is especially interested in the interaction between audio and visuals, generative systems, and shared agency between humans and machines in the artistic creation process. He is currently a PhD candidate in sound art at the University of Portsmouth (UK).

hyperobject::01 (Enrico Dorigatti & Zayd Menk)

Program Note:

Modern technology exists as a hyperobject; it is immensely vast and complex — the sheer scale of it exceeds our ability to ever fully comprehend its complexity. It is ubiquitous and in constant proliferation, encompassing vast networks of devices, software, and digital infrastructure that underpin our modern world. Through the interaction of abstract imagery and sounds rooted within post-digital and glitch aesthetics, realised through generative and algorithmic techniques, this installation explores our understanding of this complex phenomenon — our limits of reality and human perception with complex concepts such as data — inviting the audience to contemplate the vastness of the world we inhabit and our place within it as technology becomes ever more ubiquitous.

Bio:

Enrico Dorigatti is an experimental sound artist and creative technologist (see above).

Zayd Menk is a Zimbabwean interdisciplinary artist primarily working within sculpture and installation. Menk's practice attempts to dissect modern technology as a hyperobject, with inquiry into the ethical, social, and ecological implications of technological progress. Using electronic waste as his primary material, Menk creates landscapes and interactive installations. His practice has gained notable recognition, including features in CNN's Great Big Story, Aesthetica Magazine, Designboom, Hypebeast, and My Modern Met, and he was selected for the New Contemporaries 2023 program.

TINKERED THINKING / a stream of none (Sylvain Souklaye)

Program Note:

TINKERED THINKING / a stream of none is a sonic questioning and dialogue between deconstructionism and intentionality — precisely how the cultural reorganization of nature by presets and repetitive rhythms weaponizes dopamine for a collective orgy. On the other extreme of the wave, chaos emerges or follows the instinct, intense and whole, only dedicated to loneliness. The work explores the duality of these two states of being — security and freedom — as agents of social order and of natural flow.

Bio:

Sylvain Souklaye is a Brooklyn-based French multi-modal artist. He is obsessed with sampling intimacies about people who don't belong to a determinate identity, gender, class, colour or nationality. His performances are a collage of individual memories which are relived for and via the audience. Self-taught, he began performing with vandalism in Lyon, and then intimate happenings, radio experimentation and action poetry. Among his best-known pieces are *la blackline*, a 5-year durational radio performance; *le déserteur*, a digital art installation; *TME*, a docudrama performance; and *MIGRANT MARKET*.

Parramatta Resonance (Julian Knowles)

Program Note:

This site-responsive audio-visual installation listens to real-time weather data which impacts the visual and audio material in response to changing weather conditions. It explores the concept of the anthropocene and philosopher Timothy Morton's notion of the 'hyperobject' — things that are 'massively distributed in time and space relative to humans'. The work explores how creative sonification, visualisation and affective media might bring these concepts closer so we can 'feel' and experience data affectively, tapping into the micro-ecology of the Parramatta site to make the climate hyperobject seem tangible. Programmed in MaxMSP, real-time weather data is obtained through API calls to the eight closest atmospheric weather stations around the Parramatta CBD. Temperature, humidity, solar radiation, wind speed, direction and barometric pressure are used as inputs into an audio-visual compositional system.

Bio:

Julian Knowles is a composer, performer, media artist and researcher specialising in new and emerging technologies. His creative work spans composition for theatre, dance, film and television, electronic music, sound design and media arts, popular music and record production. He has created more than 50 works disseminated by high-profile record labels, broadcasters and art institutions internationally, and has been a member of the Australian electro-environmental audio group Social Interiors since the mid-1990s. His work has been presented at venues such as SFMOMA, Experimental Intermedia NYC, VIVID Sydney, and the Sydney Opera House. Julian is a Professor of Music and Media at Macquarie University in Sydney.

Concerts

Spatial Audio 1 — Choreobot: The Muse of Movement and Kinetic Expression

- Monday 9 October, 1–2pm — UNE

Timbral Shifts (Erin M Demastes) — 8 channels, 3’42”

Program Note: *Timbral Shifts* is a piece for 8-channel audio written in Csound and is an experiment in auditory perception. The piece begins with slow-moving harmonies made of difficult-to-locate sine tones and ends with the complex timbres of FM and granular synthesis that are easier to locate. As the piece progresses, the timbral shifts create a secondary layer of perceptual tension-and-release together with more traditional compositional methods such as the slow building of harmonies and textures.

Bio: Erin M Demastes is an experimental composer, performer, and instrument maker whose research combines sound and technology with humour, drama, and absurdity. She uses everyday household objects and hacked electronics for her installations and performances. Erin received an MFA in Experimental Sound Practices and Integrated Media from CalArts, a BM in jazz studies and piano performance from Loyola University New Orleans, and is pursuing a doctorate in experimental music and digital media at Louisiana State University.

N’vi’ah (João Pedro Oliveira) — 8 channels, 10’20”

Program Note: *N’vi’ah* is an Old Testament word meaning prophetess. A prophetess conveys one or more divine messages often in the form of inspired songs, and many times her words are cryptic, requiring interpretation or even translation. This work uses isolated phonemes as musical material. Intelligible words are not articulated, leaving to the listener the imagination of what their contents and meanings could be.

Bio: Composer João Pedro Oliveira holds the Corwin Endowed Chair in Composition for the University of California at Santa Barbara. He studied organ performance, composition, and architecture in Lisbon and completed a PhD in Music at Stony Brook. His music includes opera, orchestral compositions, chamber music, electroacoustic music, and experimental video. He has received over 70 international prizes and awards, including the Guggenheim Fellowship (2023), the Bourges Magisterium Prize, and the Giga-Hertz Special Award. https://jpoliveira.com

Gränstillstånd (Benjamin Carey) — 8 channels, 19’21”

Program Note: *Gränstillstånd* (Boundary States) is an octophonic fixed-media work composed during an artist residency at Elektronmusikstudion (EMS), Stockholm, in June 2023. In the author’s work with modular synthesisers, what continues to fascinate him is the unique process of human/non-human collaboration at the heart of the practice — a dynamic dance of human and non-human agencies, building Simondonian ‘open machines’. Gränstillstånd was composed from five performances recorded across a 7-day period, each in relative isolation, then composited into an overarching musical and spatial narrative.

Bio: Benjamin Carey is a Sydney-based composer, improviser and educator. He makes electronic music using the modular synthesiser, develops interactive music software and creates audio-visual works. Ben has released ten albums, including *METASTABILITY* (2023), and has collaborated with artists including Sonya Holowell, JACK Quartet, Sydney Chamber Opera, and ELISION ensemble. His work has been performed internationally at Huddersfield Festival of Contemporary Music, Sydney Festival, IRCAM Live, and VIVID Sydney. Ben is a Lecturer in Composition and Music Technology at the Sydney Conservatorium of Music.

The Unsettling Knot (Steven Ashby & Vicki Hallett) — 2/4/8 channels, 10'04"

Program Note: *The Unsettling Knot* is an inquiry into the looming question of "what's next?" as society wades through the evolving trajectory and intermingling of possibilities presented by the outbreak of COVID-19. The work develops from an interpretation of the coronavirus RNA sequence as it is allowed to mutate through a combination of stochastic and chaotic processes. With each moment, the work positions itself between an uncertain future and the myriad approaches undertaken by communities to find a way forward.

Bio: Steve Ashby is a musician, composer, and sound artist based in the United States. His work focuses on sound found in the natural and digital world to discover places of intersection which engage in the art of listening. Recent performances include ICMC 2022, Soundpedro, Moxsonic, EMS Stockholm, Radiophrenia Scotland, and NYCEMF. Vicki Hallett is a versatile musician, field recordist, sound artist, composer and educator (see UNE Installations).

Alchemy: Iron and Bronze (Ian Whillock) — 8 channels, 13'20"

Program Note: This work is a sonic and spiritual study on two materials of alchemy: iron and bronze. Throughout the piece, both instruments and non-instrumental objects constructed with these metals are used as sound sources, and all synthesized sounds are developed from their spectra and waveforms. The materials are organized in contrasting structures from singular objects to complex layered meta-objects. The work is also a study on the spirituality and symbology of the materials — used as symbols of wealth, power, violence, beauty and mysticism across civilizations, yet both decay through time with rust and patina.

Bio: Ian Whillock is a composer, multimedia artist, and audio engineer based in Austin, Texas. His works have been performed at numerous festivals such as ilSUONO Contemporary Music Week (IT), Moscow Biomechanics Festival, NYCEMF, Mise-En Festival, SPLICE, and others. In 2022 he was a resident composer at Avaloch Farms with LNK Percussion. He has won Honorable Mention in the 2023 Xenakis International Electronic Music Competition, the Ise-Shima Special Prize, and the Ida M. Vreeland Award. Ian currently teaches Audio Technology at the Art Institute of Austin.

Live Concert 1 (UNE) — Cybella: The Muse of Dreams and Virtual Reality

- Monday 9 October, 6–7.30pm — UNE

Hypnagogic Hallucination Machinery (Berk Yagli) — 8 channels, 11'40"

Program Note: *Hypnagogic Hallucination Machinery* is about the 21st-century condition. Living in a hyper-consumer-based world, where everyone happily becomes a commodity to take part in society, the concepts of individuality, freedom, privacy, and humanity once again become crucial to question and discuss. The sea of endless escapism, simultaneously fractured and monotonous people and ideas — we are now part of the systematic hallucination machine more than ever. This piece reflects these topics in an auditory way.

Bio: Berk Yagli (born 1999) is a Cypriot guitarist, composer, and producer (see Piece and Paper 3).

Foxglove (Cameron G R Naylor) — 8 channels, 15'00"

Program Note: Created from sounds of a tree isolated from its typical natural environment, *Foxglove* explores a heightened sense of scale resulting from extremely close recording techniques, reframing minuscule natural processes as functions of a great ever-growing entity. Here, almost imperceptible sounds are magnified, becoming part of a series of vast and enveloping spaces interior and exterior to the tree.

Bio: Cameron G R Naylor is an electroacoustic composer and sound artist. Through the manipulation of field recording and abstract sound material, his compositions explore sound and space as a metaphor in musical storytelling. Having completed a Masters in Electroacoustic Composition and Interactive Media at the University of Manchester in 2022, he has performed at Convergence, MANTIS, SC2022, and SOUND/IMAGE.

Waiting On The Storm To Break The Heat (Kasey LV Pocius) — 8 channels, 7'46"

Program Note: Weaving together field recordings taken during heat waves in Montreal and Cheltenham with synthesizer improvisations recorded while processing the breakdown of a romantic relationship, *waiting on the storm to break the heat* explores the dread and calm before the catastrophic, and the relief found thereafter. Through intermodulation of the synthesized elements and transformations of both materials, the summer storm breaks the heat and for a moment provides relief, allowing the listener to find a new balance in the aftermath.

Bio: Kasey LV Pocius is a gender-fluid intermedia artist originating from St. John's, Newfoundland and located in Montreal, who grew up experimenting with multimedia software while pursuing classical training in viola and piano. They are particularly interested in multichannel spatialization and group improvisatory experiences. They hold a BFA from Concordia in Electroacoustic Studies and are pursuing an MA in Music Technology at McGill under Dr Marcelo M. Wanderley.

Within these cells (Michelle Chandrajaya) — 4 channels, 8'36"

Program Note: *within these cells* (2023), for 4 speakers. When you are held captive for an indefinite amount of time, not being able to talk to others or even see the sunlight, one can only wonder what it does to a person's state of mind. The work explores this idea by expressing the paranoia and fear that a person feels as they become an inmate, living in an isolated cell block. Having to cope with post-traumatic stress disorder by themselves, alone, they slowly descend into insanity.

Bio: Michelle Chandrajaya was born in Jakarta in 2002. Music has had a dominant presence in her life since she began learning piano at age five. Moving often allowed her to listen to different types of music, and her appreciation grew. She deepened her knowledge of music theory, history, and world music while studying in the UK, then continued musical theatre and composition studies in Indonesia. She is studying at UPH's Conservatory of Music, focusing on musical composition under Dr Matius Shanboone. In 2023, her electronic music piece *within these cells* was accepted to the International Computer Music Conference in Shenzhen.

AI Phantasy (Panayiotis Kokoras) — 8 channels, 10'00"

Program Note: *AI Phantasy* was composed at the GRIS multichannel studio, the University of Montreal; the MEIT theater at CEMI, the University of North Texas; and the composer's home studio. He used a vacuum cleaner to set into vibration various membranes and other props at the end of the suction tube, modulated by hand following the Fab Synthesis paradigm, along with a series of circular pan-flute sound generators rotating electromechanically at variable speeds. The word "Phantasy" (with "Ph") refers to a state of mind of an infant child during early stages of development — constructed from internal and external reality, modified by feelings and emotions, and projected into both real and imaginary objects.

Bio: Panayiotis Kokoras is an internationally award-winning composer and computer music innovator, currently Professor of composition and CEMI director at the University of North Texas.

Hailing from Greece, he trained in classical guitar and composition in Athens and later in York, England (MA, PhD). His compositions use sound as the only structural unit, with his concept of "holophonic musical texture" describing his goal that each independent sound contributes equally to the synthesis of the total. His music calls upon a "virtuosity of sound".

Brompton & Braeswood (Timothy Roy) — 8 channels, 11'05"

Program Note: *Brompton & Braeswood* is an acousmatic piece inspired by the composer's personal experience living through Hurricane Harvey. The title derives from the intersection where he and his wife were living at the time, along Brays Bayou in Houston. Central to the piece is a library of field recordings captured at that intersection in the days immediately prior to Harvey making landfall. Having just returned from hospital after major surgery, the composer draws on contemporaneous thoughts and feelings — intense worry and gloom, but also optimism. Other sound materials include noisy tones synthesized in Max/MSP, pitched wood, and piano and guitar samples, spatialized using HOA.

Bio: Timothy Roy composes music steeped in imagery and allusion. His music has been heard in Canada, Germany, Taiwan, Japan, the UK, Chile, Cyprus, Croatia, Slovenia, and across the US, with performances at the National Theater of Taipei, Music Biennale Zagreb, ZKM Karlsruhe, BEAST, and ICMC. He has received honors from the Salvatore Martirano Award, ASCAP/SEAMUS Student Commission Award, Giga-Hertz Prize, and Prix Destellos. He is completing a doctorate at Rice University's Shepherd School of Music and is choirmaster and organist at the Church of Saint Peter in Saint Paul, Minnesota.

grind (ˈtoʊ-drɔ-ˈki) — 8.1/2 channels, 8'23"

Program Note: Each year, approximately five-hundred billion plastic cups are used, of which roughly six billion end up in landfills every year. *grind* attempts to symbolize the faulty business and political ideologies that ultimately contribute to a lack of environmental sustainability through the destructive morphology of a single sound source: a plastic Keurig coffee pod hitting the floor.

Bio: ˈtoʊ-drɔ-ˈki is a composer framing pictorial properties found in visual art — colour, shape, balance, and space — with organizing principles of philosophical, sociological, and metaphysical intent. Their music has been performed by Hypercube, Duo Sequenza, and ensemble vim, and featured at the Aspen Music Festival, SEAMUS, and ICMC. They serve as Assistant Professor of Music Technology at Johnson University and as Director of Equity, Inclusion, and Diversity for the Millennium Composers Initiative. They received their DMA in Composition from the University of Miami, Frost School of Music.

Mercury Rising (Michael Lukaszuk) — 6/2 channels, 7'04"

Program Note: *Mercury Rising* is about hearing the sounds of the 1960s space race within an acousmatic context, zooming in on the material qualities of various voices and technologies from this part of history. In addition to sourced recordings, electronic music techniques of the time are revisited with digital sampling and effects processing to consider their potential to be heard as retro-futuristic. The use of algorithmic processes to control synthesized sounds points to the possibility of AI and related forms of generativity fuelling new forms of competition and discovery.

Bio: Mike Lukaszuk is an electroacoustic composer from Kingston, Ontario. Much of his output explores the idea of generativity in music and sound art through mobile devices, musical AI and coding as artistic practice. His music has been presented at multiple ICMC events, SEAMUS, NYCEMF, ISEA, and the Australasian Computer Music Festival. He won 1st Prize in the Hugh Le Caine Awards of the 2015 SOCAN Foundation Young Composer Awards. Mike holds a DMA in Composition from the University of Cincinnati and teaches at Queen's University's Dan School of Music and Drama.

Fixed Media 1 (UNE) — KleioData: Muse of History and Data-driven Knowledge

- Tuesday 10 October, 3.40–5pm — UNE

Delusion Guitari (Roger Alsop) — 15'00"

Program Note: *Delusion Guitari* involves playing guitar into a system of eight (or more) randomly timed delays, with the signal from each delay put out to a multi-speaker system. The random delay times may be from one millisecond to 20 minutes, and the feedback for each delay is infinite. This results in an improvisation system in which every improvised action is perpetually re-played into the performance environment at an unpredictable time. For the improviser this creates a situation in which each action becomes part of each subsequent and preceding action, and as these actions concatenate and coincide, a sonic environment is built.

Bio: Roger Alsop is a multidisciplinary artist who focuses primarily on collaborative and improvised artworks (see Paper Session 2A).

0285e52e (Tiernan W Cross) — 4'39"

Program Note: *0285e52e* is a data-driven composition created by combining over one thousand short field recordings recorded by the composer and arranged based upon their ranging characteristics of spectromorphological complexity. Edited using statistical analysis, this piece plays on the embellishment of spectral fluctuation, spatial motion, timbre, dynamic range and differences in recording quality.

Bio: Tiernan W Cross is an electroacoustic composer and researcher based in Adelaide, Australia. His work focuses on utilising statistical analysis and data science to inform compositional decisions. His work has been exhibited throughout Asia, Europe, North America and Australia (including CTM Festival, Cannes Film Festival, MONOM, and The Spatial Sound Institute), with commissions by Screen Australia, Screen NSW and the ABC. His research has been published by Intellect, the Technoetic Arts Journal, the Orpheus Instituut, AES and the Electroacoustic Music Society.

sin-título[08-12-22] (Roy Guzmán) — 11'55"

Program Note: This is an acousmatic work generated with algorithms. Structurally mesoscopic musical events were composed and permuted. The algorithms are studies from sources the composer enjoys the most, from Afro-Cuban rumba to complete serialist works, as well as the use of geometrical forms for durational and frequential data. This piece uses only instruments of Puerto Rican music like the panderos (hand drums), the cuatro (a Puerto Rican string instrument), and the güiro.

Bio: Roy Fernando Guzmán Rodríguez (b. 1987) is an experimental, contemporary, and algorithmic instrumental and electronic music composer, sound artist, improviser, and poet born in San Juan, Puerto Rico. His music is primarily exploratory and uses algorithmic procedures to create musical structures. He is a founding member of the Puerto Rican experimental folkloric trio Abolengo and founder of CMEPR (Colectivo de Música Experimental de Puerto Rico). He has performed and presented works across the Netherlands, Slovakia, Hungary, Austria, the US, Canada, Argentina, Chile and Mexico.

Elegy 1: Time Present is Time Past (Irving P Kinnersley) — 14’59”

Program Note: *Elegy 1* was composed during the first year of COVID, using sound recorded before the pandemic around the city of Wells in Somerset, England (the twelfth-century cathedral clock, the cathedral bells and the sounds of a domestic garden). Traces of memory and the past mingle with a disturbing present in a composition which, like English poetry’s elegiac tradition, is haunted by the passing of time and mortality. Bells sound throughout the composition, and processed bell sounds become an imaginative response to the times when the bells have fallen silent, as during the Second World War and the COVID pandemic.

Bio: Irving P Kinnersley holds a degree in English Literature and Religion from the University of London, a Master’s in Modern Literature from the University of Kent and an MMus in Creative Music Technology from Bath Spa University. He is studying for a PhD at the University of Manchester. His work has been selected for the MANCA Festival, Seeing Sound, Festival Ecos Urbanos, Festival DME, Audio Mostly, the MANTIS Festival, and the Convergence Festival. *Elegy 1* was awarded first prize in the student category of the Klang International Electroacoustic Competition.

Musical Journey Around the World (Mirjana Dokic) — 3’27”

Program Note: This project is created in Max/MSP and Ableton Live. Without drastic changes to the original material, it expands beyond traditional musical forms, investigating the creative possibilities of multichannel signal processing for generating rich soundscapes. It processes the sound of percussion instruments (Macedonian drum “tapan” and tarabuka) and voice using a soundscape generator where a single sample is the starting point, with panning algorithms ranging from multichannel LFOs to playback-status-driven techniques. The second part uses multichannel harmonic delay for viola sound processing, superimposition, and all-pass filtering of 50 delayed copies of the same sample.

Bio: Mirjana Dokic is a composer, new media artist and violist based in Hong Kong, where she is pursuing a PhD in Creative Media at City University of Hong Kong. She holds a Master’s in Music (Viola) from the Conservatory of Music Santa Cecilia, Rome, and a Bachelor’s from University Ss. Cyril and Methodius, Skopje. As a fusion violist and sound artist, Mirjana expands the role of her instrument beyond classical music into electroacoustic music. In 2018 she released her debut album *Beautiful Baby* under the artist name Mia Del Sol.

Ossa (Paolo Montella) — 7’02”

Program Note: “In this world which we enter, appearing from a nowhere, and from which we disappear into a nowhere, Being and Appearing coincide” (Hannah Arendt). Field recording, as an operation determined by rituals, times, and techniques, releases the need to be artistically varied. In this work, field recording is grafted onto the practice of dozens of interviews conducted between June and December 2021 with women from small towns in southern Italy. Agata, an 11-year-old girl, shows us her field at the centre of tense and contradictory forces, between her being a child and wanting to become a woman.

Bio: Paolo Montella is an electroacoustic composer, multi-instrumentalist, and programmer. Field recording and radical improvisation practices are central to his aesthetic, and he focuses his research on the relationship between sound and source. He graduated in Electronic Music at the Naples Conservatory with M° Elio Martusciello. His works have been performed at festivals such as Sonic Cartography (Chatham, UK), XIII CIM (Ancona), Interferenze, Supersonique Festival (Marseille), and Tempo Reale (Firenze).

Temporary Residing (Fulya Uçanok) — 7’02”

Program Note: *Temporary Residing* explores a myriad of moments, and the movements that bridge them. These movements are sometimes enacted through sudden ruptures and at others through subtle leakages and transitions. The piece invites listeners to move through various residences, temporarily inhabiting each space while exploring diverse sound behaviours and listening situations. It is messy, it is plural, it is always situated.

Bio: Fulya Uçanok is an electroacoustic musician and pianist — composing, performing and improvising. Born in Turkey, she studied classical piano performance at Hacettepe University Ankara State Conservatory and completed her master’s at MİAM, Istanbul Technical University, then completed a doctorate focusing on electroacoustic music composition and performance. Her PhD dissertation is titled *Towards a Response-able Com-position Practice*. She has a duo (iKKi Duo) with cellist Zeynep Ayşe Hatipoğlu, is a member of klank.ist ensemble, and co-initiated Soundinit.

Riverside AV Works 1 — Calligrapix: The Muse of Calligraphy and Visual Arts

- Tuesday 10 October, 10.30am–12pm — Riverside Theatres

Xeno (Enrico Dorigatti) — Fixed AV, 3’43”

Program Note: *Xeno* is a multimedia work based on the intense, dense, and continuous twine between audio and video. ‘Xeno’, a Greek word indicating something extraneous, describes this work as it proposes a novel, uncommon, and subverted relationship between the two media. Breaking the rule that sonic events are always the consequence of physical actions, Xeno proposes an alien reality in which this relationship is altered, making it difficult to understand which of the two media is the cause or the consequence of the other. The meaning also refers to the rapidly, continuously mutating multimedia content as a whole.

Bio: Enrico Dorigatti is an experimental sound artist and creative technologist (see UNE Installations).

Hearing Earth I — Sober (Gabriel Zalles) — Stereo AV fixed, 5’07”

Program Note: *Hearing Earth* is a project which seeks to turn mineral data acquired by scientists at Scripps Institute of Oceanography into musical material by quantizing element signatures, acquired via spectroscopy, to musical scales. 188 images were scraped from an online database and digitized, then post-processed in MATLAB to isolate the peaks; the compressed information was fed into Max/Pd to create chords, then imported into a DAW to compose *Sober*, a haunting AV piece reminding us of our role in the ecosystem and our responsibility to future generations. This piece is a collaboration with Lei Liang and Emily Chin at UCSD.

Bio: Gabriel Zalles Ballivian is a Bolivian student living in San Diego and attending UCSD, where he is working towards a PhD in Computer Music. He is an audio engineer specializing in immersive audio, specifically ambisonics, with experience in MATLAB, JUCE (C++), and open-source hardware. His research and artwork have been shown at NIME, AES, NYCEMF, SMC, and LAC.

Remapping Earth (Cecilia Suhr) — 11’46”

Program Note: *Remapping Earth* is a real-time interactive audio-visual video art exploring the sound of various ecosystems, including machine and human, existing simultaneously. In doing so, it visually represents the new terrain of sonic mapping.

Bio: Cecilia Suhr is an intermedia artist and researcher, multi-instrumentalist (violin/cello/voice/piano/bamboo flute), multimedia composer, interaction designer, painter, author, and improviser. She has won multiple awards including the MacArthur Foundation Digital Media and Learning Research Grant (2012) and the Pauline Oliveros Award from the IAWM (2022). Her music has been performed at NYCEMF, ICMC, SEAMUS, and many others. She is the author of *Social Media and Music* (2012) and *Evaluation and Credentialing in Digital Music Communities* (MIT Press, 2014), and is Associate Professor in the Department of Humanities and Creative Arts at Miami University Regionals.

Luar (Nicola Fumo Frattegiani) — Stereo AV fixed, 9'32"

Program Note: *LUAR* is the inexorable descent of the man within himself — a fall towards the most hidden places of his soul where the protagonist struggles with the multiple representations of the self. The poetry and words of poet Fernando Pessoa accompany this descent towards never-ending subjectivity. LUAR is a Portuguese word meaning 'moonlight', the quintessential nocturnal light, the one and only light amongst the shadows, which allows one to see the hidden side.

Bio: Nicola Fumo Frattegiani is an electroacoustic and audio-visual composer living in Perugia, Italy. His research deals with electroacoustic music, sound for images, video, art exhibitions and compositions for theatrical performances. He is a Subject Expert in "Electroacoustic" and "Computer Music" at the Conservatory of Music of Perugia, held the chair of Electroacoustic Music Composition at the Conservatory of Music of Messina, and is currently professor of Sound Design at the Academy of Fine Arts in Macerata.

Estuaries 4 (Bret Battey) — Stereo fixed AV, 8'45"

Program Note: *Estuaries 4* is the fourth and final part of the author's *Estuaries* audio-visual series. It explores contrasts between intense and frenetic textures and a gentler poetics, the latter expressed in part through visualisation of the mathematical Rosenbrock function. The series involves visualizing Nelder-Mead optimization, a process used to find solutions to complex, multi-variable problems. The music was created with the author's Nodewebba software, which interlinks pattern generators to create complex emergent behaviours.

Bio: Bret Battey is Professor of Audiovisual Composition at the Music, Technology, and Innovation Institute for Sonic Creativity at De Montfort University, Leicester, UK. He creates electronic, acoustic, and audiovisual concert works and installations with a focus on generative techniques. He has been a Fulbright Fellow to India and a MacDowell Colony Fellow, and has received recognitions from Prix Ars Electronica, Bourges, Punto y Raya, and the International Computer Music Festival.

Oceana (Amanda Stuart) — 9'48"

Program Note: *Oceana* (2023) was created as part of the Polar Sounds project, a collaboration between Cities and Memory, the Helmholtz Institute for Functional Marine Biodiversity and the Alfred Wegener Institute. The aim was to highlight the soundscapes of Antarctica, the importance of sound communication between animals and how it is affected by climate change. The two-minute field recording chosen was a mixture of killer whale and Ross seal sounds (whistles, pulsed calls and echolocation clicks) recorded with hydrophones. The whole soundscape was composed from this one sample, creating an otherworldly, augmented polyphony. The visuals were created from fourteen of the composer's paintings, transformed, layered and synced to the soundtrack.

Bio: Amanda Stuart is a composer and multimedia artist. She creates soundworlds and landscapes designed to take you on an immersive journey full of emotion and drama. She was awarded her MMus in Creative Music Technology from the Royal Welsh College of Music & Drama with Distinction. Her piece *Song of the Trees* won the IAWM Pauline Oliveros Prize for Electroacoustic Media 2015. Performances include Earth Day Art Model, NYCEMF, Boston New Music Initiative, ICMC (Perth), and SMC/SMAC (Stockholm).

b_k_n ( ) (Nicola Cappelletti) — 2/4 channels, sound only, 8’55”

Program Note: *b_k_n( )* is the representation of a generating force, an explosive event that produces the beginning of life in a timbral and structural dimension where the contrast between stillness and dynamism creates a formal balance poised between possibility and impossibility. The overall circular trend offers interrupted paths of ascent and descent, led by timbral research made with the preparation of traditional instruments such as violin and electric bass, to generate sound materials feeding electronic synthesis processes.

Bio: Nicola Cappelletti (ITA/FRA) is an electroacoustic sound artist, performer and composer. After studying violin he graduated in electronic music at the F. Morlacchi Conservatory in Perugia. Winner of the XV National Prize of the Arts (Electroacoustic Composition), his research focuses on the relationship between acoustic sound and electronic treatment in relation to audiovisual works, theatre and contemporary poetry. His work has been presented at ICMC, SMC, NYCEMF, and others. He is a member of the Opificio Sonoro ensemble.

Aurora (Donna Hewitt, John Montgomery & Stéphanie Pouliquin) — AV, 4’10”

Program Note: *Aurora* explores the dormant Dionysian essence within each of us. As night descends, it awakens our primal instincts, an irresistible force that takes hold of us, bringing forth both chaos and transformative new order. This work invites the audience to contemplate the innate power and potential of embracing our wild and untamed nature. The work incorporates AI images made in collaboration with Midjourney.

Bio: Stéphanie Pouliquin is a French-Australian teacher of French language and culture at Alliance Française of Canberra, with a passionate interest in philosophy, literature, arts, and poetry inspired by nature and Greek mythology. John Montgomery is a playwright, actor, composer, performer, producer, video artist and educator with a particular interest in merging artforms; he is the Curriculum Inspector – Creative Arts for the NSW Education Standards Authority. Donna Hewitt is a performer, vocalist, electronic music composer and instrument designer, inventor of the eMic, and Associate Professor in Music at the University of New England. https://donnahewitt.net/

Riverside AV Works 2 — Lyrictron: The Muse of Music Production, Poetry and Phonemes

- Tuesday 10 October, 2–3.30pm — Riverside Theatres

VoceST II (Massimo Fragalà) — Stereo fixed media, 3’40”

Program Note: All the sounds that form this composition derive from the elaboration of a vocal sample. Starting from a very small sample (0.1756 seconds), the composer tried to change the original characteristics in order to generate a range of sounds more or less different from their original variety, using sound processing techniques such as time and spectrum stretching, morphing, harmonisation, freezing and sustaining a sound on an explicitly specified grain, and transposing copies of sound on top of one another.

Bio: Massimo Fragalà graduated in Electronic Music and in Classical Guitar. His music has been performed at many festivals and conferences worldwide including ICMC (2003, 2005, 2006, 2018, 2019), LAC, Emufest, Vox Novus 60x60, Csound Conferences, WOCMAT 2016, MusLab, and NYCEMF 2022. He was awarded 3rd Prize ex aequo in the 4° Concorso Internazionale di Composizione Elettronica "P. Schaeffer" (Italy, 2003).

Desolation (Andrea Montanaro) — Stereo fixed, 8'31"

Program Note: *Desolation* is an acousmatic composition made with digital and environmental sound materials. The title is inspired by the desolation inherent in a man who does not accept belonging to the social fabric he lives in because he does not recognize himself in it. Digital synthesis is obtained through subtractive synthesis processes: given a sound source rich in harmonics, it is filtered from a spectral point of view by "subtracting" frequency bands or single partials. This filtering develops during the temporal evolution of the signal, organized into three sound typologies (low-, mid-, and high-pitched) and rich in beats.

Bio: Andrea Montanaro is a multi-instrumentalist composer, director and teacher from Abruzzo. He obtained diplomas in popular music, clarinet, piano, jazz piano, organ and organ composition, multimedia composition, and film music composition at the Conservatories of Pescara, Rome and Verona. He collaborates with international artists such as Franco D'Andrea, Calogero Palermo, and Xavier Girotto, and teaches Analysis and Composition Theory and Musical Technologies at the Liceo Musicale of Vasto.

String Theory (Mark Oliveiro) — live electronics & Paetzold contrabass recorder, 9'00"

Program Note: *String Theory* is an interactive concert work that intertwines the resonant tones of the Paetzold contrabass recorder, live electronic music featuring sub-bass frequencies, and immersive video projections inspired by Galileo Galilei's celestial research. Designed using MaxMSP, this experience takes the audience on a journey through the harmonious realms of both music and astronomy. Through MaxMSP, the recorder's melodies and textures blend seamlessly with live electronic manipulations, blurring the boundaries between the earthly and the ethereal. *(To be performed by Alana Blackburn.)*

Bio: Dr Mark Oliveiro is a graduate and distinguished alumnus of the Sydney Conservatorium, the University of North Texas and the School of Music of Indiana University (see Paper Session 2B).

Whispers of the Branch (Cecilia Suhr) — AV stereo fixed, 8'17"

Program Note: *Whispers of Branches* is an improvisational violin and bamboo flute work on fixed and live electronics, performed as a real-time audio-visual interaction. The improvisational sound of the violin and bamboo flute is processed electronically while affecting the drawing lines of the branches. This piece imagines the whispers of branches after all the leaves have fallen off. Can there be a message of fortitude and strength as branches withstand the cold months of winter?

Bio: Cecilia Suhr is an intermedia artist and researcher (see Riverside AV Works 1).

Discontinuous Mediation III (Rodrigo Pascale) — 5.1, 5'53"

Program Note: *Discontinuous Mediation III* is the third composition of this series. Unlike the first and second, which used pre-recorded guitar and clarinet sounds, here pre-recorded viola sounds are used as sonic material. As in the predecessors, the sounds were organized in a segmented way, forming a structure based on the composer's compositional procedure. This method explores the conceptual idea of a perceptually continuous world being interpreted by a discrete mediation.

Bio: Rodrigo Pascale (b. 1996) is a Brazilian composer based in the USA. He graduated in composition from UFRJ in 2018 and is pursuing a DMA in composition at Peabody Institute. His works have been selected for festivals including Panorama da Música Brasileira Atual, MUSLAB 2020, Espacios Sonoros 2020, and the Tesselat Electronic/Electroacoustic Concert. He won the International Composition Contest organized by Sound Silence Thought and the Festival Expresiones Contemporáneas 2020.

Action-Reaction (Chi Wang) — AV work, 8'30"

Program Note: In classical mechanics, Newton's third law states that for every action, there is an equal and opposite reaction. In this composition, the GameTrak's retractable tethers interact with the performer's push, pull, release and free movements. Each time the performer releases the tethers, they are retracted to the rest state, creating a predictable yet unique realignment path. The data measured from the physical movements are mapped to control various parameters, creating musical expressions that are both superimposed and nuanced. The performer, sound-producing engine, and the concert venue can be virtually connected from three different physical locations, producing the composition via network.

Bio: Chi Wang is a composer and performer of electroacoustic music. Her interests include sound design, data-driven instrument creation, composition, and performance. Her compositions have been performed internationally at ICMC, NIME, Musicacoustica-Beijing, SEAMUS, NYCEMF, Kyma International Sound Symposia, and WOCMAT. Her compositions were selected for the SEAMUS CD and Best Composition from the Americas from ICMC. She received her DMA at the University of Oregon and is currently an assistant professor of music (composition: electronic and computer music) at the Indiana University Jacobs School of Music.

Astraeus (Eric Lemmon) — AV work stereo, 6'52"

Program Note: Astraeus // Consort of Eos // God of the Dusk.

Bio: Composer Eric Lemmon's artistic practice and academic research is preoccupied with the politics that circumscribe and are woven into our musical technologies and institutions. His music has been reviewed by the New York Times and featured on WQXR's Q2, and has been performed in venues ranging from (le) Poisson Rouge and SubCulture to the DiMenna Center for Classical Music. Works by Eric have been performed internationally, with performers including PUBLIQuartet, Jacqueline LeClaire, and Yarn/Wire. His music has been performed at SEAMUS, PMF∼, the Xenakis Networked Performance Marathon, the NowNet Arts Conference, the Australasian Computer Music Conference, the Web Audio Conference, and NYCEMF.

The Swan [Derbarl Yerrigan] (2022) (Sze Tsang) — AV work, 19'38"

Program Note: *The Swan [Derbarl Yerrigan]* (2022) was originally created as part of a series of works commissioned by the State Library of Western Australia and West Australian Music, re-imagining the State Library's maps collection. The work combines a map of the Perth area from the 1890s, rendered into a digital virtual instrument, with field recordings taken from modern-day Perth, demonstrating how landscape can be both inspiration and active participant in a composition.

Bio: Sze Tsang (they/them) is a published researcher, performer and audio-visual artist residing in Boorloo/Perth, Western Australia. Sze's practice and research focus on the relationships between self and place through audio-visual works. Their practice involves incorporating audio and visual elements of place into compositions, as an expression of catharsis through abstracted soundworks. Sze performs and exhibits works nationally and internationally as samarobryn, and is a PhD candidate at the Western Australian Academy of Performing Arts, Edith Cowan University.

Live Concert Riverside — EuterMechania: Muse of Music Interfaces and Instrumentation

- Tuesday 10 October, 6–7.30pm — Riverside Theatres

Parakletos (Patrick Hartono) — fixed AV work, 10'24"

Program Note: *Parakletos* is an interactive audiovisual composition that represents the initial outcome of an ongoing experiment investigating the relationship between sound and visuals. "Parakletos" is a Koine Greek word meaning advocate or companion. The individual roles of the sound and visual elements complement each other, presented through a compositional approach that allows for real-time interactivity. A custom interactive music system enables both elements to communicate dynamically based on the performer's decisions. The sound component serves as the fundamental structure, while the visual element is relatively free, its original form altered in real-time while still being continuously bound with sound through data derived from spectrum analysis.

Bio: Patrick Gunawan Hartono is an electroacoustic composer and audiovisual artist, born in Makassar, Indonesia, in 1988. He is known for his innovative integration of traditional Indonesian music, computer-generated sounds and visuals, and field recordings. He graduated Cum Laude from the Rotterdam Conservatory (with study at the Institute of Sonology), holds a Master of Music in Sonic Arts from Goldsmiths, University of London, and undertook a Live Electronic Course at IRCAM. His work has been showcased at ICMC, ACMC, YCMF, WOCMAT, NIME, NYCEMF, and Ars Electronica. He is working towards a doctoral degree at the University of Melbourne.

Sonic Mutations (Alexis Weaver & Rowan Savage) — live, 15'00"

Program Note: *Work 1: Blakly We Cry (Wagahn)* explores how technology allows a Kombumerri man to deepen and express his relationship with crows (wagahn) through an AI trained on both his field recordings of crows and on his own voice. We hear a live transformation from human to crow and back again, against a backdrop of AI-generated field recordings. *Work 2: Control Yourself* opens on a simple, sung 'prompt' melody and quickly branches off into various sonic scenes, guided by the AI's melodic response. The tool (affectionately dubbed 'AI-Lexis') blurts out unintelligible words — visceral utterances which convey emotion beyond human language.

Bio: Rowan Savage (salllvage) is a proud Kombumerri man, living on Wangal Land, working at the intersection of queer club culture and connection with Country (see Paper Session 3B). Alexis Weaver is an electroacoustic composer based in Sydney, Australia (see Paper Session 2C).

Musing on Computer Labour (Niamh Dolfi-McCool) — live

Program Note: *Musing on Computer Labour* focuses on the invisible efforts and strains that the average laptop undergoes. It's a reflection and experimentation on how users interact with technology. Using Pure Data and Python, the piece tracks the live CPU usage data of the laptop, which is fed into a reactive soundscape. As more windows are opened and videos played, the Pure Data synths swell and drum patterns become more distorted. The interface is the laptop itself: at first responsive, but as CPU usage rises, the laptop-as-interface struggles — the mouse drags and it reacts erratically.

Bio: Niamh is a multi-disciplinary artist currently based in Narrm/Melbourne. Niamh is all about accessibility of education, particularly anything related to tech and creativity. By day, Niamh works as a tutor at ANU, teaching creative computing to university and high school students; by night, she co-runs the DJ collective Vessel, organising workshops and events. She strongly believes in shifting computing to have care, empathy and culture at its core.

Kind regards. For a friend (Anna Savery) — live, 5'00"

Program Note: This piece explores approaches to audio-visual compositional processes within the intermedia space. A 2D animation of a bird is given agency through an audible voice and interacts with the human performer (a violinist with a sensor-enhanced bow) by using that voice in a call-and-response style, as well as playfully navigating flight paths and wing movements. Focusing on relationships and memories, the bird and the human performer move through a journey of initial tentativeness, joyful familiarity and a slightly afflicted, chaotic resolution.

Bio: Anna Savery is a violinist, composer, improviser and music technologist (see Paper Session 2A).

Breathing 1 (Sophie Rose) — live, 6'25"

Program Note: *Breathing I* uses Bilateral Coordination as a movement framework for live voice, electronics (wearable gesture control), and projections. The work uses Ableton Live 11 and Max, exploring panic by manipulating live and sampled breath. It uses asymmetrical movement derived from a children's game melded with a therapeutic parallel. There is one live voice, one granularised breath, and one breath in a self-built tape-scrub Max-for-Live plugin. The left arm scrubs audio horizontally and the right arm scrubs vertically. Gesture functions as grounding through intense concentration on being in the present.

Bio: Sophie Rose is a multi-instrumentalist, singer, composer, improviser, researcher and multimedia artist undertaking a PhD at The University of Melbourne in Interactive Composition (see Paper Session 2A).

A Way to the Dungeon (Xin Lu & Yuchen Li) — live AV with Max/MSP and Ableton, 5'01"

Program Note: In this computer-based musical creation inspired by RPG games, a symphony of chiptune echoes resounds, inviting listeners to embark on a wistful voyage evoking emotions both nostalgic and new. Its melodies dance with playful energy, evoking memories of heroic battles, treacherous dungeons, and magical encounters. The vibrant tones of synthesized instruments, reminiscent of the 8-bit era, interweave to form intricate harmonies while embracing modern production techniques. The work uses Max MSP and Ableton Live for the live performance, distributing workflows between the two with Max for Live, and OBS Studio for recording.

Bio: Xin Lu is an undergraduate student at The Australian National University in the third year of a Bachelor of Advanced Computing (Research & Development), interested in computer science research and music. Yuchen Li is an undergraduate student at ANU in the Honours year of a Bachelor of Arts in linguistics, interested in acoustic phonetics and music.

Innerlogue (Rita Yung) — Fixed stereo, 8'50"

Program Note: *Innerlogue* is a dialogue inside a person's mind. Everyone has more than one personality, and these personalities appear in different places, times, and situations. Innerlogue uses different timbres, sounds, and articulations of the bassoon to represent multiple personas and the conversations between these characters, and uses live electronics to expand each characteristic.

Bio: Rita Yung is pursuing her Master's degree in music technology and digital media at the University of Toronto. She received her Master's in composition at the University of Texas at Austin and her Bachelor's at Hong Kong Baptist University. As a composer, Yung works on both acoustic and electronic music. Her works have been featured at the 69th International Rostrum of Composers (2023), ICMC (2017 Shanghai, 2020 Chile), Electric LaTex, Etchings Festival, IAWM, and Seoul International Computer Music Festival.

Locked Groove 4 Mix (Michael Callander) — modular DJ set, 3'30"

Program Note: 51 instances of audio created with Ableton Live's Operator synthesizer, each recorded for 1.8 seconds at 133.33 bpm and organised in Live's Session View as looping Audio Clips. Follow Actions were used to generate random playback at various intervals. The original Audio Track was duplicated multiple times, and LFOs applied to the volume fader for each track to impart an additional degree of chance upon the composition.

Bio: Dr Mike Callander is a Lecturer in Music Industry at RMIT University (see Paper Session 1A).

Spatial Audio 2 — Astraline: The Muse of Astronomy, Space and Acoustics

- Wednesday 11 October, 1.10–2.40pm — UNE

Analogue Revolutions versus Digital Strings (Nicole L Carroll & John Ferguson) — 2 channels (live analogue synths), 10'00"

Program Note: Each performer works with DIY and self-built systems. Through semi-structured gestural improvisation, the musical agenda focuses on highlighting analogue/digital characteristics and celebrating similarity/difference. *Analogue Revolutions* features variable step-sequencers driven by two independent clocks controlled via potentiometers/photocells, each outputting to an individual voltage-controlled oscillator, with two distortion circuits and touch-points bringing wild, grungy tones. The digital performance system is built in Max/MSP and includes physical modelling, granular sampling, and feedback loop components. The goal is to celebrate analogue/digital characteristics while traversing noise aesthetics, contemporary computer music, and the repetitive rhythms of groove-based pop forms.

Bio: Dr John Robert Ferguson is a post-digital/electronic musician and sound/multimedia artist based in Brisbane, Head of Creative Music Technology and Senior Lecturer at Queensland Conservatorium Griffith University. He builds and performs electronic instruments and post-digital systems that foreground tactile interaction. His work has been presented at ICMC, NIME, and the Australasian Computer Music Conference, among many others. https://johnrobertferguson.com/

Dr Nicole L Carroll is a composer, performer, sound designer, and builder working with audio, video, and tangible objects, across noise, soundscape, and acousmatic genres. She performs electronic music under the alias "n0izmkr" and builds custom synthesizers, controllers, and performance sensor systems. Her works have been performed internationally at SEAMUS, ICMC, TEI, and NIME. She holds an MA and PhD in Computer Music and Multimedia from Brown University and is Senior Lecturer in Music at Queensland University of Technology.

Night (Allison Ogden) — 2/4/6 channels, 8'54"

Program Note: *Night*, for computer-generated sound, with or without female dancer, was written using Max/MSP at The Computer Music Studio at The University of Chicago and premiered at Stanford University. It has since been choreographed by Alycia Scott of Columbia College Chicago and later by Korina Kordova of the Silesian Dance Theater, where it was incorporated into the contemporary dance show *Karolina*.

Bio: Dr Allison Ogden works as an Assistant Professor of Composition and Literature at the University of Louisville. She has a PhD from The University of Chicago, has written a number of pieces of music, climbed many mountains and hiked many trails, and brought two human beings into this world.

125±10Hz x4 (Fern Bravo) — 2/4 channels, 10'49"

Program Note: This piece explores auditory phenomena through multiple sine tones clashing with each other across the frequency spectrum and stereo field. These are then sent through tonal and phase-shifting effects and returned to sum together.

Bio: fern is a multidisciplinary sound artist/producer based in Tāmaki Makaurau. Their work explores sound production and the external experience of tactile composition and tone shaping, in the melding of signals interacting through hardware and software.

Nophesis (Serkan Sevilgen) — 5'05"

Program Note: The sound materials for the piece are recorded sounds from Istanbul. Two sets of recordings — birds and human-made noises — hint at the different nature of sounds we hear in the city. The audio samples are matched randomly to create pairs. Spectral analysis and resynthesis tools in Csound are used to generate cross-synthesized sounds where the amplitude and frequency values from each audio are mashed together.

Bio: Serkan Sevilgen is an electroacoustic music composer who employs his professional programming skills to create computer music. He works as the Director of Engineering of the catastrophe modeling company Temblor, Inc. He has an MA in Sonic Arts from MIAM at Istanbul Technical University. He is a co-founder of Soundinit and a founder of WORC, a networked music ensemble, and is a member of the Istanbul Coding Ensemble. His work has been presented at ICMC, NIME, SMC, ISMIR, and NYCEMF.

Post Anthropocene Record No. I (Edmar Soria) — 8 channels, 8'23"

Program Note: In 1950, Alan Turing published "Computing machinery and intelligence", proposing the "imitation game" and beginning with the question: can a machine think? Seventy years later, a new question is proposed: can a machine dream? This acousmatic work represents (from a speculative point of view) the experiential record of the first machine that was able to dream, some time ahead in an unknown future, placing us at the first-person perspective of its first dream.

Bio: Edmar Soria holds a PhD and Master's in Music Technology from UNAM, a Bachelor's in Mathematics from the National Polytechnic Institute, and a Master's in Applied Economics from UNAM. He is a research professor in the Department of Arts and Humanities at the Universidad Autónoma Metropolitana – Unidad Lerma, Mexico, where he heads the research area PiATS. He has won or placed in the Acousmonium INA GRM competition, the Iannis Xenakis International Electroacoustic Contest, and SONOM 2014, and has undertaken residencies at Musique & Recherches (Belgium), INA GRM (Paris), DXArts, and CMMAS (Mexico).

In Threaded Memory (Vicki Hallett & Steve Ashby) — 8/2 channels, 7'31"

Program Note: *In Threaded Memory* focuses on the transients of memory as it drifts from present to recollection. With the movement of time, older events mingle amongst a continually updated catalogue of new experience. Through generative lines of delay and feedback, the iterations of clarinet and guitar synthesize a perception of memory drift and remembrance of space, finding the listener ensconced in a cycle of reflection oscillating between presence, remembering, and reconsidering.

Bio: Steve Ashby is a musician, composer, and sound artist based in the United States (see Spatial Audio 1). Vicki Hallett is a versatile musician, field recordist, sound artist, composer and educator (see UNE Installations).

HYPNOS (Domenico De Simone) — 2/4/8 channels, 7'01"

Program Note: Hypnos and Thanatos — Sleep and Death. Death mirrors Sleep, because it is the latter that interacts with life; it is life itself, while Death represents its mirror opposite. Here Hypnos is introduced; Thanatos can wait. Starting from sounds recorded on the lakeside of the town where the composer was born, he imagined what the soundscape will be in the future.

Bio: Domenico De Simone is Professor of Electroacoustic Composition at the "Umberto Giordano" Music Conservatory of Foggia. He graduated in Piano, Jazz, Composition and Electronic Music, and completed advanced study at the Accademia Nazionale di Santa Cecilia under Azio Corghi and in Electronic Music under Giorgio Nottoli. He attended the Accademia Chigiana in Siena, receiving diplomas of merit in Music for Film (Ennio Morricone) and Composition (Franco Donatoni). His compositions have been performed in more than one hundred concerts in Italy and abroad and broadcast by RADIOTRE.

A Different Perfect (Neil Crispin Aldridge) — 8 channels, 4'32"

Program Note: *A Different Perfect* is a journey through a realm where technology and creativity converge. The interplay between manipulated vocals, guitars, and electronic instrumentation serves as a conduit for emotion and imagined narratives that transcend conventional boundaries.

Bio: Neil Crispin Aldridge is an innovative New Zealand-based sound artist. With a background in music production and film sound, he creates intricate, intimate soundscapes that confuse distinctions between composition and sound design. His work has been exhibited internationally, most recently in the French pavilion at the Venice Biennale of Architecture.

Fixed Media 2 (UNE) — CybErato: The Muse of Love, Sensors and Sound Synthesis

- Wednesday 11 October, 10.30am–12pm — 28 George St, Parramatta

Exploring The Technological Muse of the Roland TB-303 Bassline Synthesizer (David Haberfeld / Honeysmack) — live web performance, 15'00"

Program Note: The performance explores the iconic TB-303 Bassline Synthesizer and its various clones through an improvised all-hardware live electronic dance music (EDM) performance. It presents real-time sonic examples born from the unique TB-303 sound and demonstrates why it has been a significant contributor to the sonic development of EDM since Acid House. Roland's TB-303 was manufactured as an electronic bass accompaniment machine but proved difficult to program and was discontinued in the early 1980s. Later in the decade, DJs picked up the now-inexpensive 303 and, manipulating its limited real-time controls, produced a squelching, bubbling bass timbre — the sonic meaning of Acid House was born. As much as Rock 'n' Roll owes its existence to the electric guitar, EDM found its electric guitar in the TB-303.

Bio: Honeysmack (David Haberfeld) is an electronic musician from Melbourne, Australia. He has been producing music and performing live Techno for almost three decades, earning a reputation as one of Australia's leading electronic music pioneers. Honeysmack's music grew from the underground rave scene of the early 90s, combining Acid and Techno. He has released numerous albums and EPs, toured extensively and completed a PhD in Acid Techno. His live performances are known for their high-energy, improvisational nature, incorporating hardware synthesizers and drum machines — no laptops, no records, just pure live electricity.

Der hohle Zahn (Nicola Fumo Frattegiani) — Stereo, 4'44"

Program Note: *Der hohle Zahn* is inspired by Wim Wenders' *Wings of Desire* (1987). "Der hohle Zahn" (the hollow tooth) is how Berliners nicknamed the Kaiser-Wilhelm-Gedächtniskirche church after it was bombed in 1943. From the ruins of the bell tower, Damiel the angel observes the movement of human life. The composition presents exclusively acoustic materials of flugelhorn: the natural sound symbolises the vitality and conflictual nature of humanity, while manipulated sounds represent the gaze of the angel. The incipit and conclusion overlap — the circle is born and dies within the same instant and at the same point.

Bio: Nicola Fumo Frattegiani is an electroacoustic and audio-visual composer living in Perugia, Italy (see Riverside AV Works 1).

Papua Acousmatic Sound (Markus Rumbino) — Stereo, 6'00"

Program Note: The destruction of tropical forests in Papua threatens the natural sounds that live in the region; exploitation of Papua's natural resources continues to this day. *Papua Acousmatic Sound* is a reinterpretation of musical experience formed through personal and social cultural collective memory in Papua.

Bio: Markus Rumbino is a composer, sound artist and researcher who uses sound material recorded from the daily activities of human life and the sounds of nature. He studied music composition at the Indonesian Institute of Art Yogyakarta, completing a Master of Music, and is active as a music instructor at the Indonesian Institute of Cultural Arts Tanah Papua. He was selected for the Cultural Activist program in New Zealand (2016), attended the Europalia Festival with the Voice of Papua music group (2017), and founded the Alyakha Art Center in Yokiwa village.

Entomon study (Ian Stevenson) — 8 channels, 3'30"

Program Note: This work imagines a post-singularity scenario in which machines are left to contend with the insects, which probably pre-date humans by 300–400 million years. Inspired by Michel Serres' eco-philosophical *Biogea* and our nation's yearning to embrace First Nations spirituality, insect life suggests both the alien and the familiar. The work combines physically modelled acoustic insect simulations with algorithmic behaviour patterns in 8-channel spatial audio.

Bio: Ian Stevenson is a specialist in the field of audible design with over twenty years of experience as an audio engineer, producer, artist and educator. He is currently senior lecturer in music and sound design at the University of Technology Sydney. His current research is in spatial audio, sound design, sound studies, soundscape analysis, and music and sound pedagogy.

I am Not My Body (Donna Hewitt & The House) — 8 channels, 4'10"

Program Note: *I am Not My Body* is a new choral work created with all-female choir 'The House', intended to form part of a new electronic AI opera called *Femina Controllata Machina* (Woman Controls Machine).

Bio: Donna Hewitt is a performer, vocalist, electronic music composer and instrument designer, inventor of the eMic, and Associate Professor in Music at the University of New England. https://donnahewitt.net/

The Unbearable Lightness of Words (Elsa Jaeyoung Park) — 2–6 channels (web), 21'26"

Program Note: If we could measure the weight of words — be it positive or negative — and that measurement was audible, how would we react to one another? *The Unbearable Lightness of Words* is an interactive, web-based sound cinema installation exploring 'one's sound of mind', based on sonification of a South-Korean family story with violent language and psychological data of Korean adults' language experiences in 2019–2022. It is a sound-documentary recording the collective memory of people in Korea today, as well as exploring an emotionally communicative channel where listeners can share their own 'weight of words'.

Bio: Elsa Jaeyoung Park (1991) is a composer based in South Korea. She studied electroacoustic music at the University of Birmingham and Jazz Piano at Seoul Institute of the Arts. Her main interests focus on exploring uncomfortable emotions in the composition of electroacoustic music and data sonification. Her music has been heard in the USA, Australia, UK, Germany and South Korea, featured at ICMC, SICMF, Telematic Festival, and ACMC. She won the best piece of ICMA 2018 regional award Asia-Oceanic and the best sonification piece at the Coastal Futures Ecoacoustic Music Competition in 2021.

破境 [pò jìng] // beyond realms (Yichen Wang & Sandy Ma) — 4 channels AV live, 10'00"

Program Note: 破 [pò] – to break, to burst, to fracture; 境 [jìng] – boundary, border. Mixed reality treads the boundary between the physical and digital realms. 破境 muses upon the spaces on either side of the boundary, investigating the symbiosis formed by mutual language. The performance is a 10-minute improvised dialogue between an augmented-reality (AR) enabled instrument and a traditional Volca synthesiser, exploring the tensions and connections in mixed reality interactions. As computer-controlled lighting changes, players on both instruments respond to the colour change to find harmony and foster sonic communication.

Bio: Yichen (逸宸) Wang (she/her) is a PhD researcher in the Sound, Music and Creative Computing lab at the Australian National University. Her research combines augmented reality and human-computer interaction with artistic practice in new musical interface and expression design. She has performed with her mixed reality musical instrument *isometric-sen* since 2022, including at OzCHI2022 and the NIME2023 music track. Sandy Ma (she/her) is an emerging multidisciplinary artist with a background in ceramics and creative computing, exploring interaction and tactility through the intersections between computing, sculpture, and sound. She is working and studying on Ngunnawal/Ngambri country.

InVoce (Kai R Dreyfus-Ballesi / The Binary Beats Brigade) — AV smartphone collaboration, 6'12"

Program Note: *InVoce* is a computer choral performance that delves into the expressive qualities of the human voice through the integration of a motion and gesture interface. This live performance involves an ensemble of smartphone musicians, each controlling an expressive vocal synthesizer, resulting in a collaborative and improvised exploration of melody and harmony.

InVoce extends beyond the limitations of traditional vocal capabilities, allowing performers to augment their ordinary utterances and venture into new realms of vocal expression through nuanced, expressive gestural control.

Bio: The Binary Beats Brigade consists of four computer musicians from the Australian National University Laptop Ensemble. A multidisciplinary group of computer science and music students, the Brigade creates dynamic computer music through a combination of live coding and novel musical interfaces.

Lanes Concert — Techalia: The Muse of Technology & Invention

- Wednesday 11 October, 5–8.30pm — Car Park behind 28 George St Parramatta (next to Eat St Car Park), part of the Parramatta Lanes Festival

Electronica Bass (Mansell Laidler) — one channel, 30 mins

Program Note: Welcome to *Electronica Bass*, where the electric bass takes centre stage as a powerful interface for the creation of electronic music. Throughout this performance, the artist harnesses the boundless potential of the electric bass, moulding its raw sound through a carefully curated collection of effects and digital amplifier models. Using the versatile software looping platform Ableton, the processed bass parts are looped, building intricate layers of sonic richness that defy convention.

Bio: Mansell Laidler is a seasoned bass player with a 25-year career in the music industry, including a decade as a member of the Australian Army Band. In 2020, during the pandemic, he discovered the world of Live Looping and software looping, combining the electric bass with effects and software-based loopers. He was accepted into UNE's Master of Philosophy program in May 2021, undertaking a research project titled "Electronica Bass: The Electric Bass as an Interface for the creation of Electronic Music."

String Theory (Mark Oliveiro) — live electronics & Paetzold contrabass recorder, 9'00"

Program Note: *String Theory* is an interactive concert work that intertwines the resonant tones of the Paetzold contrabass recorder, live electronic music featuring sub-bass frequencies, and immersive video projections inspired by Galileo Galilei's celestial research (see Riverside AV Works 2).

Bio: Dr Mark Oliveiro is a composer and academic lecturer at AIM (see Paper Session 2B).

Earcandles (Rodney Berry) — Stereo, 20 mins

Program Note: It's a guy with a modular synth on a guitar strap making droney ambient noise music using a mixture of digital and analogue sources and processors.

Bio: Rodney bought a second-hand Atari ST in 1989 and became a computer musician. Before that, he made do with drainpipes and pieces of wood. Perhaps best known for his installation works exploring artificial life and augmented reality technologies, lately he is enjoying live performance with a wearable modular synthesiser as well as designing affordable microtonal instruments.

Fox n Drum (Jon Drummond) — 15–30 mins

Program Note: Modular synthesisers intertwine with the enchanting sounds of acoustic sounds and instruments. #FoxAndDrum #ModularSynthesis #KochiBells #LiveStudioEP #SonicJourney #MusicalExploration

Bio: https://www.newcastle.edu.au/profile/jon-drummond · https://www.foxcontrolmusic.com/bio

Nik Nak (Nick Wishart) — 20 mins

Program Note: NikNaK creates electronic music and immersive live visuals using Round Sounds, a motion-controlled instrument. With a practice that jumps freely between creative coding and circuit bending, ṈìkṈáK is also known as the evil genius behind the Sydney-based cult band Toydeath.

Bio: Nick Wishart is an interdisciplinary artist and musician with a focus on light, sound and interactivity. Over the past 25 years, his practice has revolved around writing, playing, and recording music for live theatre and moving image, alongside video production, coding, and embedded electronics. He develops wireless, MIDI-controlled instruments using gyroscopes and accelerometers to translate human movement into sound and music — and to control DMX lighting, video, and VR. As a founding member of Toydeath (1995), he has performed at the Sydney Biennale, Big Day Out, and The Great Escape Festival, and toured Europe, the USA, and Asia.

Lowkey Austin presents Superette (Lachlan Austin) — Stereo, 30 mins

Program Note: Artist: Lowkey Austin (Lachlan Austin). Songs: *89 Cressida*, *Take the Ride*, *NewTech*, *Show me Luv*, and *Stadius* — all by Lachlan Austin.

Bio: Lowkey Austin has played his unique cruise-faced glitch electronica at notable venues such as Oxford Art Factory and Knox Bar Underground, supporting groups such as Spirals and Ears. After a few years' hiatus living in Asia, he's back with more energy and punch. A graduate of SCU Contemporary Music, he has extensive experience playing in bands and producing.

Performances

Transient (Jingyin He)

- 28 George St, Ground Floor

Program Note:

transient is a human-mecha performance featuring three human performers with mechatronic sound machines (MSMs). The MSMs and their sonic manifestations draw upon the concept of ensemble and instrumentality in their development. Musically, the piece focuses on the *jeu* of each mechatronic sound machine (via material and motion) — reflecting and exploring the sonic interactions among themselves and with the environment. Each MSM focuses on a unique sound-producing mechanism: *kinetic.bowls* involves the creation of spatial trajectories with custom-made kinetic loudspeakers; *spectra* uses motion and deformation to explore audiovisual relationships; and *stratum* utilises a rotating acrylic disc and solenoid to render audible the latent sonority of a planar material object.

Bio:

Blake Johnston is a sound artist, technologist, academic, and composer from Aotearoa, New Zealand, whose practice sits at the intersection of experience design and emerging forms of technology. Bridget Johnson creates immersive sound installations and performances that heighten the audience's experience and perception of space, developing new interfaces for real-time spatialisation. Jon He (Jingyin He) is an experimental sound and integrated media artist and researcher who draws upon ancient Chinese sound practice and emerging technologies, re-visioning them as gestural controllers, interactive systems and mechatronic musical instruments for installations and live performances.

SHOALZ: Sound in Colour (Matt Bray)

- UNE Concert Space, 28 George St
- Wednesday 11 October, 4–4.30pm

Program Note:

SOUND IN COLOUR is a 30-minute live Telematic Music Performance (TMP) between London (UK) and either Perth (Australia) or on location at the conference. Two musicians located continents apart co-create a ‘comprovised’ Electro Dub piece that also generates an immersive, reactive visual feed viewable through a VR headset or an online stream. The ethos of the piece is to demonstrate the capacity to create immersive, paradigm-shifting artworks together despite disparate locations. Telemidi is a targeted approach to MIDI network design with an explicit aim to minimise the obstruction of latency within live TMP events across a Wide Area Network. Telemidi networks exchange only MIDI information to benefit from the relatively miniature data packet sizes and the multifaceted capacity of this digital protocol.

Bio:

Matt Bray has operated as a professional musician (drummer, singer, guitarist, MIDI performer) since 1996, playing over 3,300 gigs and composing over 170 musical pieces (listed with APRA), including a piece performed by the Melbourne Symphony Orchestra and song-writing credits on an Australian top-40 album and top-20 single. Using MIDI technologies in live environments since 1998, Matt developed methods for incorporating sample- and synth-based performance in live performance, leading to his 2018 Masters research demonstrating how MIDI can be exchanged over the internet to facilitate real-time performance from any location. He is completing a PhD at the Western Australian Academy of Performing Arts (WAAPA), researching further measures to reduce network latency and present Network Music Performances within virtual environments.